

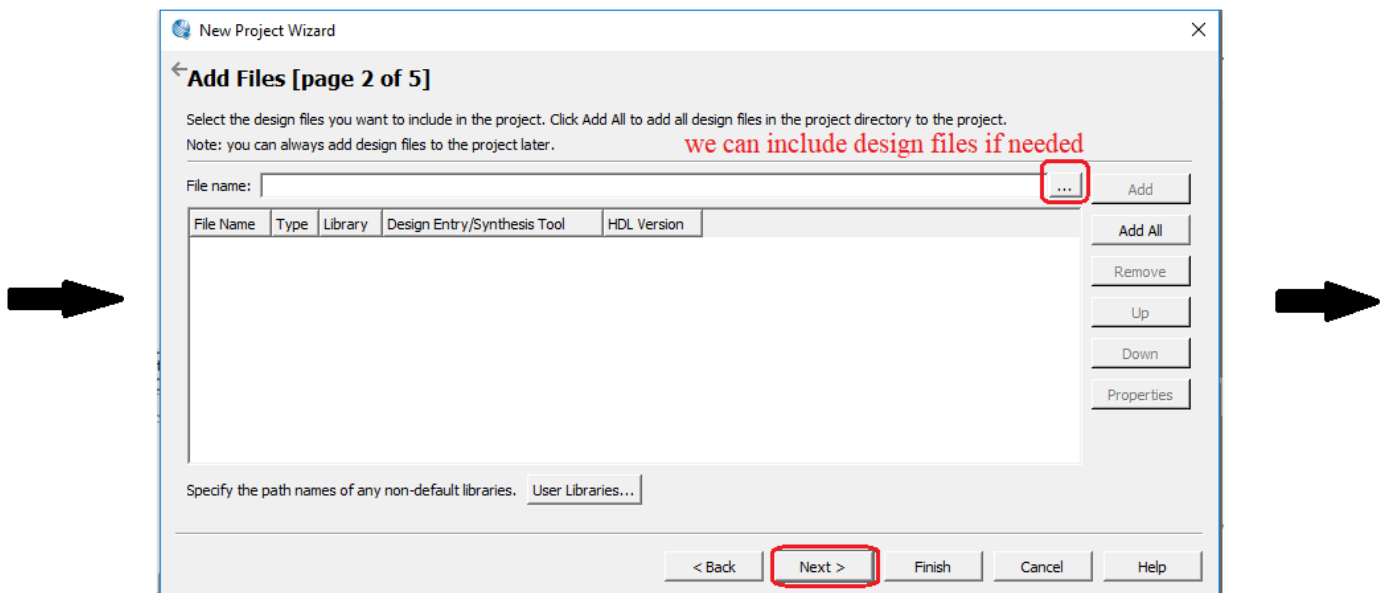
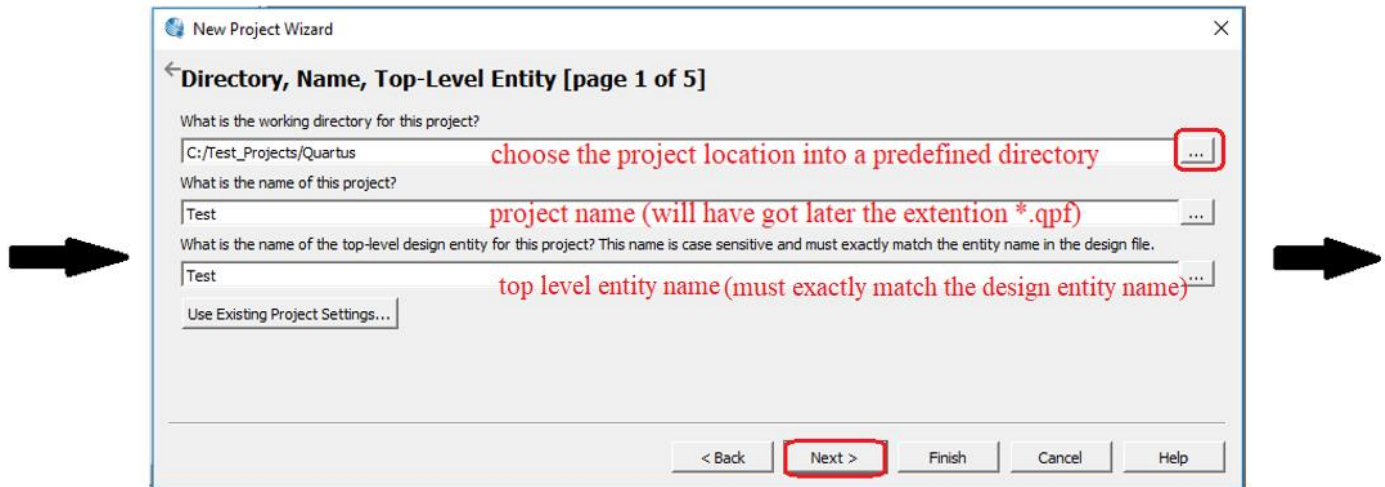
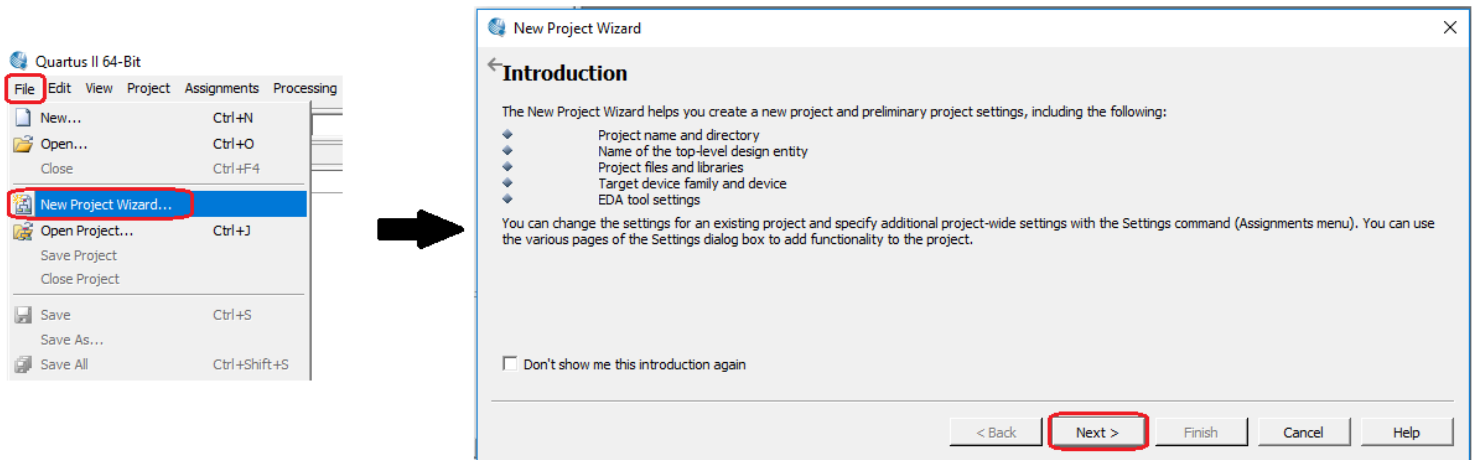
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Quartus - Create or Open a Project

A. Create a New Project:

1. Step 1 – create a new project and add VHDL files:



New Project Wizard

Family & Device Settings [page 3 of 5]

Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.

To determine the version of the Quartus Prime software in which your target device is supported, refer to the [Device Support List](#) webpage.

Device family

Family: Cyclone IV E

Device: All

Target device

☐ Auto device selected by the Filter

☒ Specific device selected in 'Available devices' list

☐ Other: n/a

Show in 'Available devices' list

Package: Any

Pin count: Any

Core speed grade: Any

Name filter:

☒ Show advanced devices

Device and Pin Options...

Available devices:

Name	Core Voltage	LEs	Total I/Os	GPIOs	Memory Bits	Embedded multiplier 9-bit elements	PLLs	Global Clocks
EP4CE115F23C9L	1.0V	114480	281	281	3981312	532	4	20
EP4CE115F23I7	1.2V	114480	281	281	3981312	532	4	20
EP4CE115F23I8L	1.0V	114480	281	281	3981312	532	4	20
EP4CE115F29C7	1.2V	114480	529	529	3981312	532	4	20
EP4CE115F29C8	1.2V	114480	529	529	3981312	532	4	20
EP4CE115F29C8L	1.0V	114480	529	529	3981312	532	4	20
EP4CE115F29C9L	1.0V	114480	529	529	3981312	532	4	20
EP4CE115F29I7	1.2V	114480	529	529	3981312	532	4	20
EP4CE115F29I8L	1.0V	114480	529	529	3981312	532	4	20

< Back Next > Finish Cancel Help

New Project Wizard

EDA Tool Settings [page 4 of 5]

Specify the other EDA tools used with the Quartus II software to develop your project.

EDA tools:

Tool Type	Tool Name	Format(s)	Run Tool Automatically
Design Entry/Synthesis	<None>	<None>	☐ Run this tool automatically to synthesize the current design
Simulation	ModelSim-Altera	VHDL	☐ Run gate-level simulation automatically after compilation
Formal Verification	<None>	<None>	
Board-Level	Timing	<None>	
	Symbol	<None>	
	Signal Integrity	<None>	
	Boundary Scan	<None>	

< Back Next > Finish Cancel Help

New Project Wizard

Summary [page 5 of 5]

When you click Finish, the project will be created with the following settings:

Project directory: C:/Test_Projects/Quartus

Project name: Test

Top-level design entity: Test

Number of files added: 0

Number of user libraries added: 0

Device assignments:

Family name: Cyclone IV E

Device: EP4CE115F29C7

EDA tools:

Design entry/synthesis: <None> (<None>)

Simulation: ModelSim-Altera (VHDL)

Timing analysis: 0

Operating conditions:

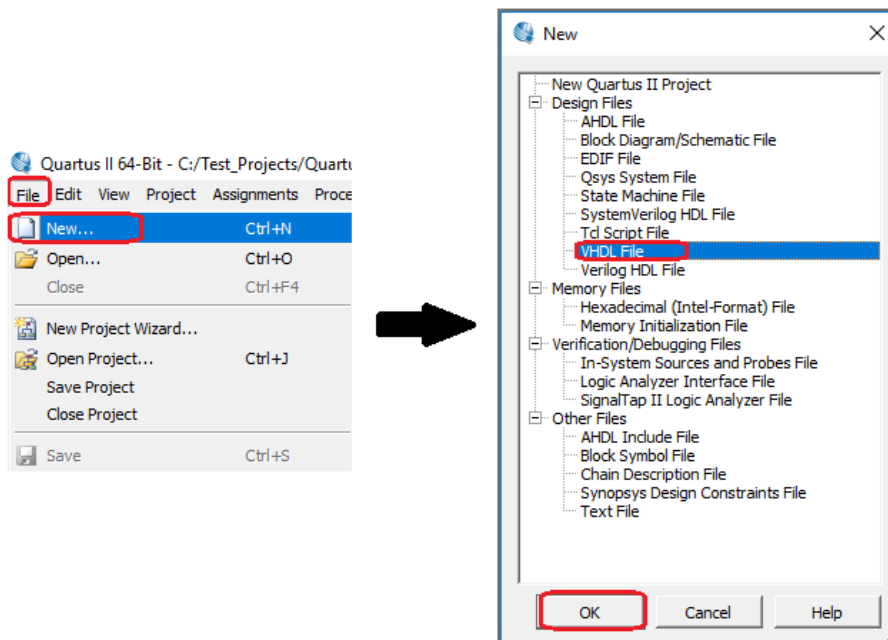
Core voltage: 1.2V

Junction temperature range: 0-85 °C

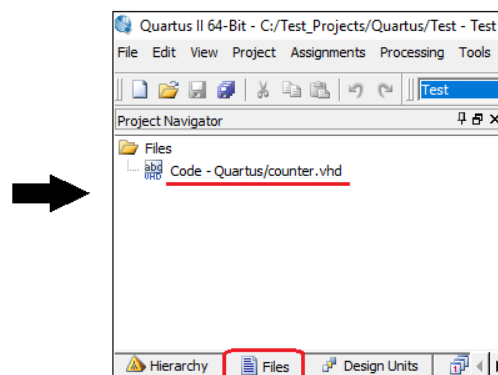
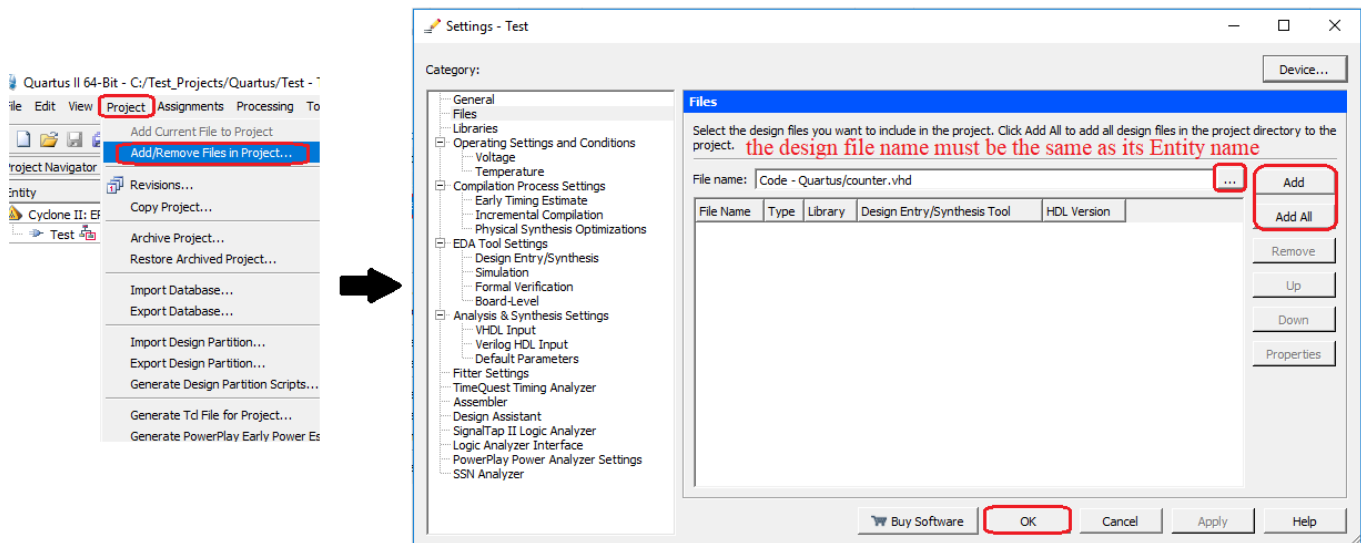
< Back Next > Finish Cancel Help

2. Step 2 – Add project files:

- In order to open a VHDL blank file, use the next step (if you're using VHDL existing files, copy these files into the project folder and skip to clause b):

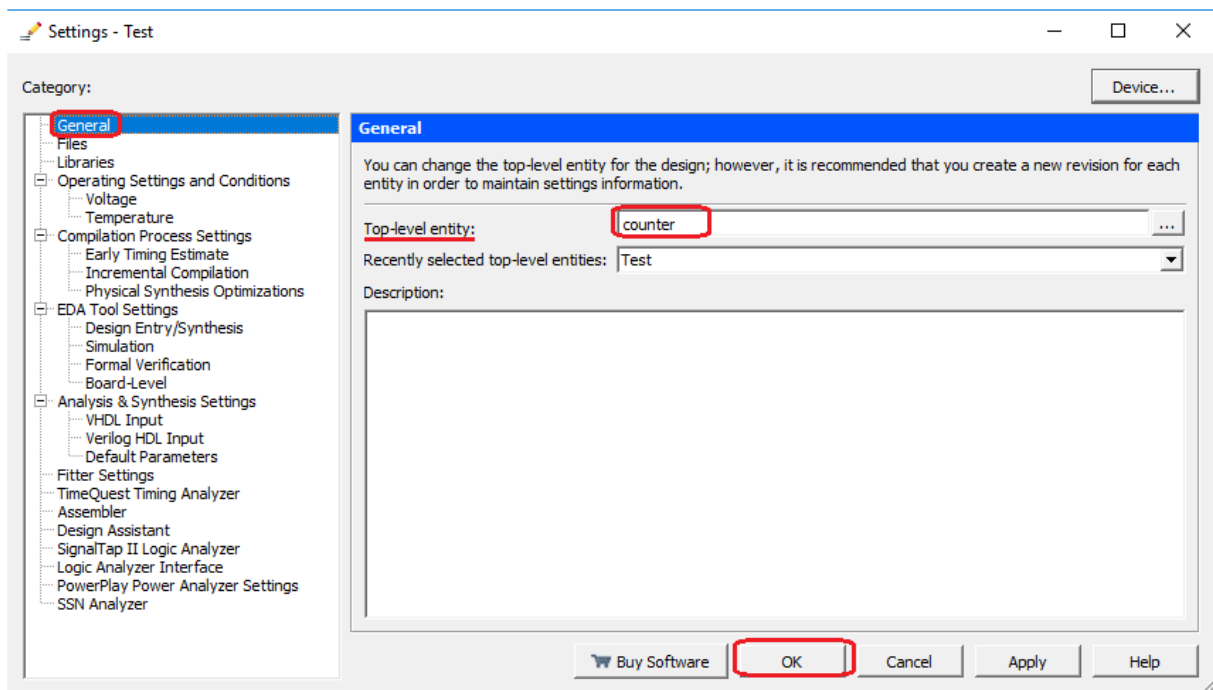
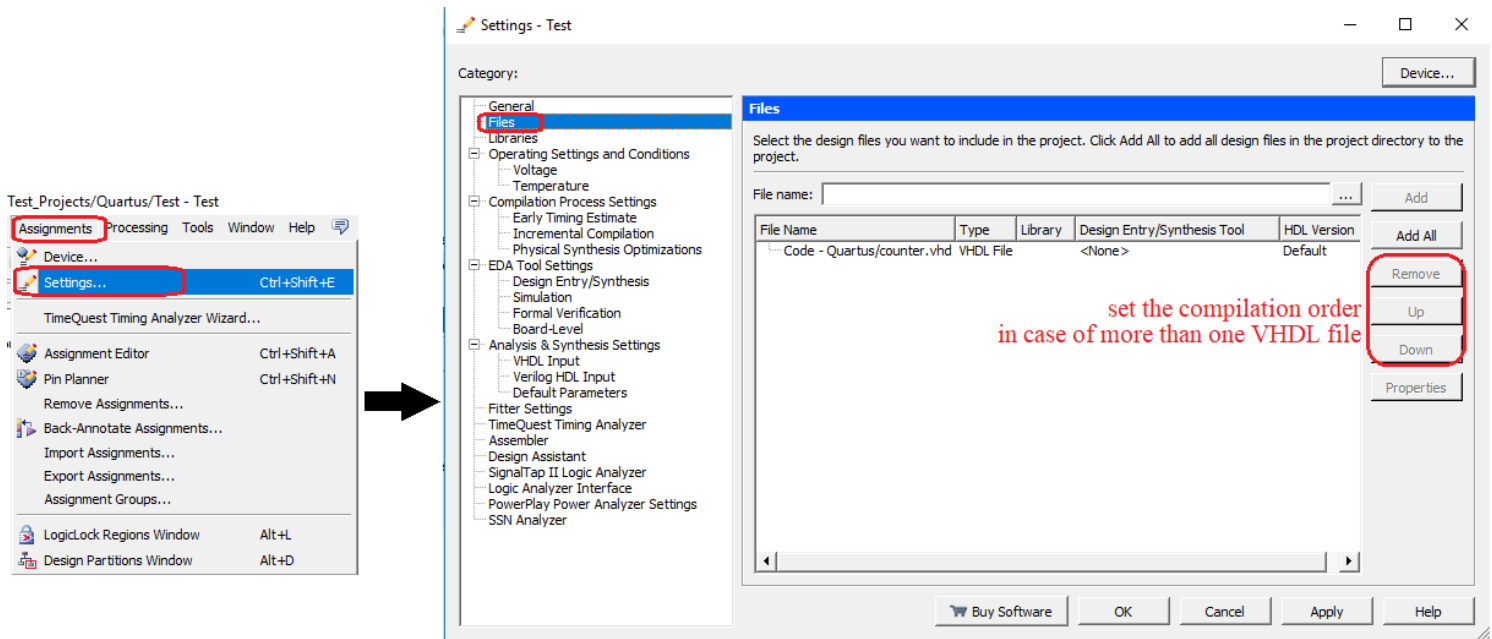


- Add the project VHDL existing files:



3. Step 3 – Code compilation:

a. Set the compilation order and top level entity:



counter.vhd

```
library ieee;
use ieee.std_logic_1164.all;
use IEEE.std_logic_unsigned.all;

entity counter is port (
    clk,enable : in std_logic;
    q          : out std_logic_vector (7 downto 0));
end counter;

architecture rtl of counter is
    signal q_int : std_logic_vector (31 downto 0) :=x"00000000";
begin
    process (clk)
    begin
        if (rising_edge(clk)) then
            if enable = '1' then
                q_int <= q_int + 1;
            end if;
        end if;
    end process;
    q <= q_int(31 downto 24); -- Output only 8MSB
end rtl;
```

Example application:

- 32bit behavioral counter with enable
- 8-bit MSB of counter output are connected to red LEDs (LEDR7-LEDR0)
- Enable connected to switch SW0
- 50MHz on board clock named CLOCK_50

b. Code compilation:

Quartus II 64-bit - C:/Test_Projects/Quartus/Test - Test

File Edit View Project Assignments Processing Tools Window Help

Project Navigator

Files

- Code - Quartus/counter.vhd

Tasks

Flow: Compilation Customize...

Task	Progress
Compile Design	00
Analysis & Synthesis	00
Fitter (Place & Route)	00
Assembler (Generate programming files)	00
TimeQuest Timing Analysis	00
EDA Netlist Writer	00
Program Device (Open Programmer)	00

Compilation Report - Test

Flow Summary

Flow Status: Successful - Tue Mar 26 15:04:19 2019

Quartus II 64-bit Version: 12.1 Build 177 11/07/2012 SJ Web Edition

Revision Name: Test

Top-level Entity Name: counter

Family: Cyclone IV E

Device: EP4CE115F29C7

Timing Models: Final

Total logic elements: 32 / 18,752 (< 1 %)

Total combinational functions: 32 / 18,752 (< 1 %)

Dedicated logic registers: 32 / 18,752 (< 1 %)

Total registers: 32

Total pins: 10 / 315 (3 %)

Total virtual pins: 0

Total memory bits: 0 / 239,616 (0 %)

Embedded Multiplier 9-bit elements: 0 / 52 (0 %)

Total PLLs: 0 / 4 (0 %)

Quartus II

Full Compilation was successful (11 warnings)

OK

c. Start analysis and synthesis:

File Edit View Project Assignments Processing Tools Window Help

Compilation Report - Test

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis

Flow Summary

Flow Status: Successful - Tue Mar 26 15:13:14 2019

Quartus II 64-bit Version: 12.1 Build 177 11/07/2012 SJ Web Edition

Revision Name: Test

Top-level Entity Name: counter

Family: Cyclone IV E

Device: EP4CE115F29C7

Timing Models: Final

Total logic elements: 32

Total combinational functions: 32

Dedicated logic registers: 32

Total registers: 32

Total pins: 10

Total virtual pins: 0

Total memory bits: 0

Embedded Multiplier 9-bit elements: 0

Total PLLs: 0

Start Analysis & Synthesis

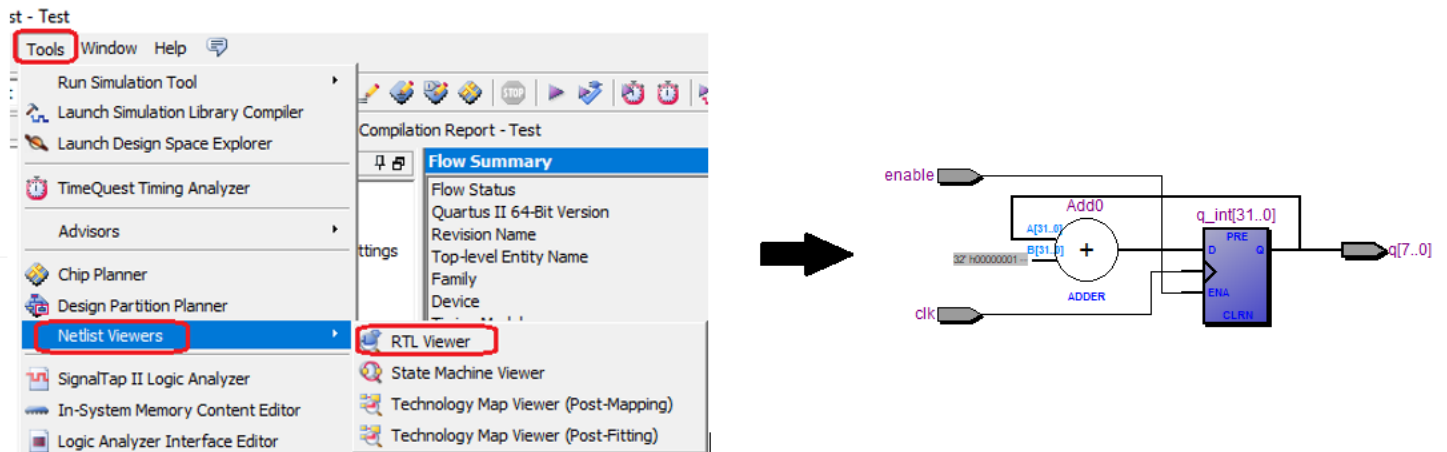
Quartus II

Analysis & Synthesis was successful (1 warning)

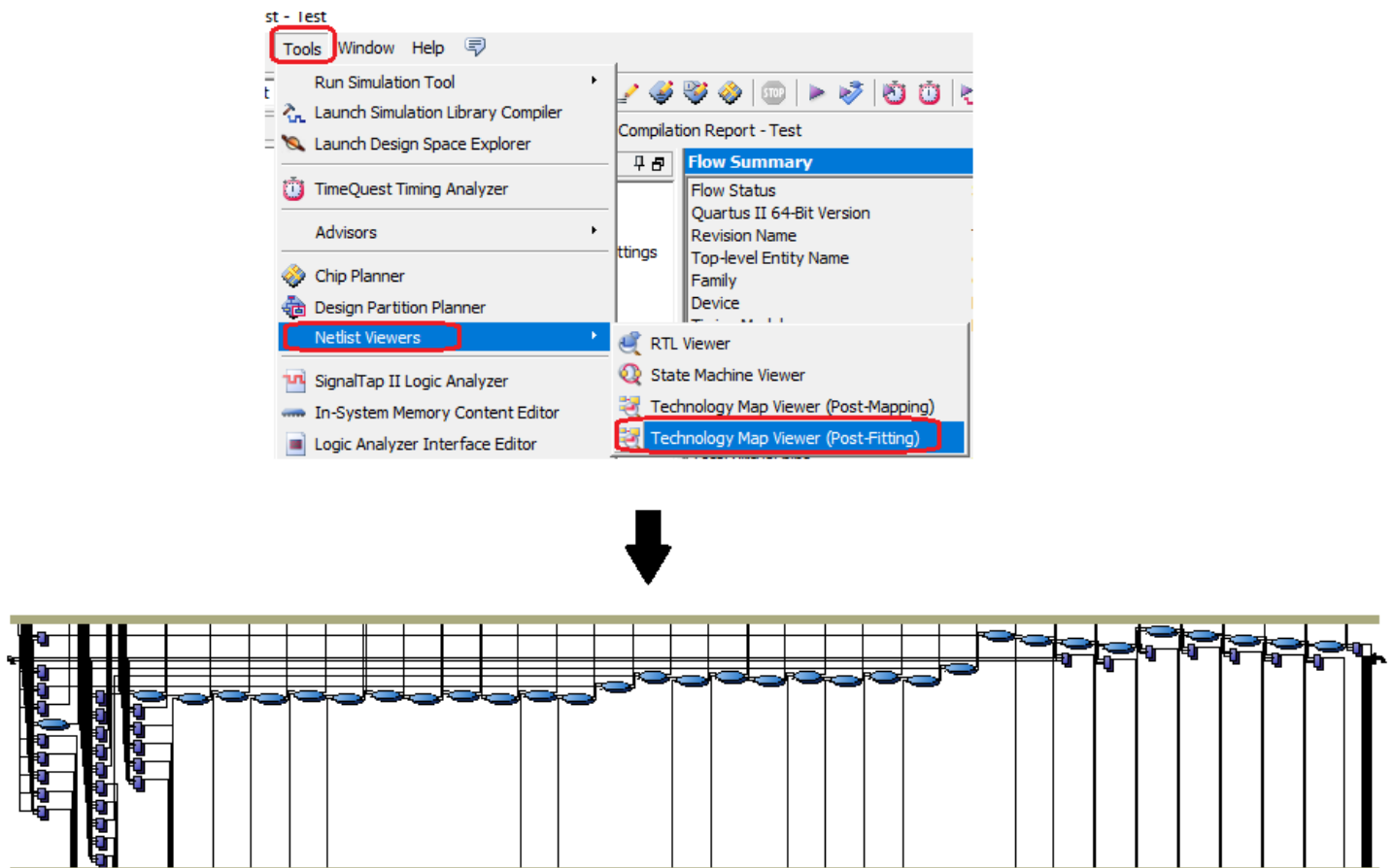
OK

4. Step 4 – Synthesis results:

a. Synthesis RTL viewer:

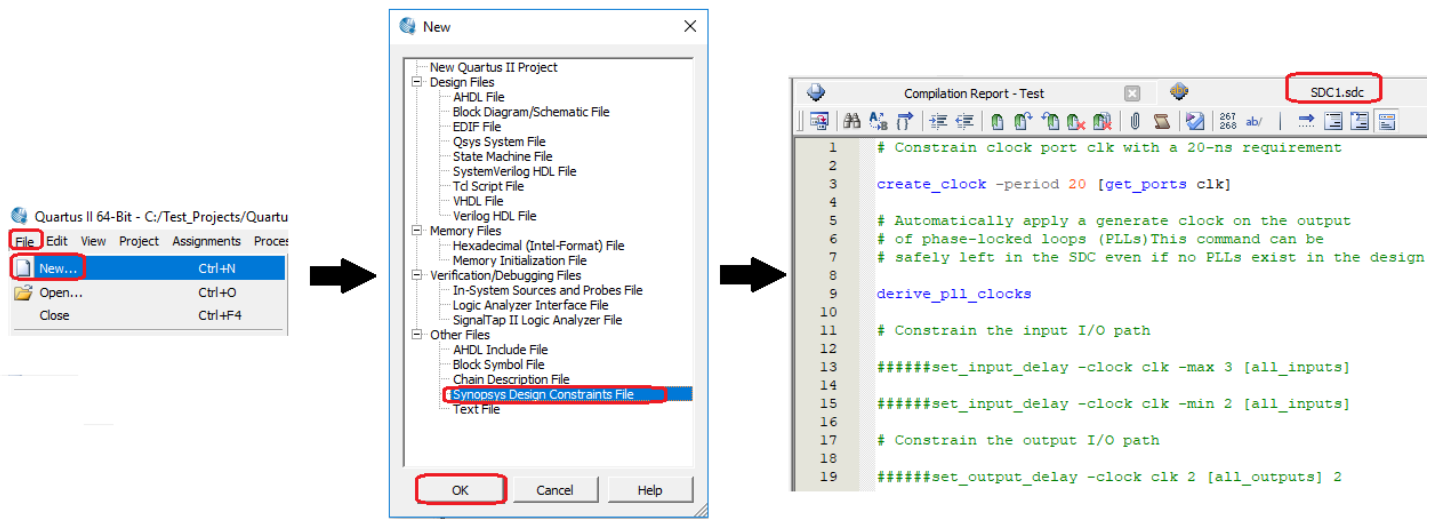


b. Synthesis Map (Post-Fitting) viewer (LEs and FFs combination):

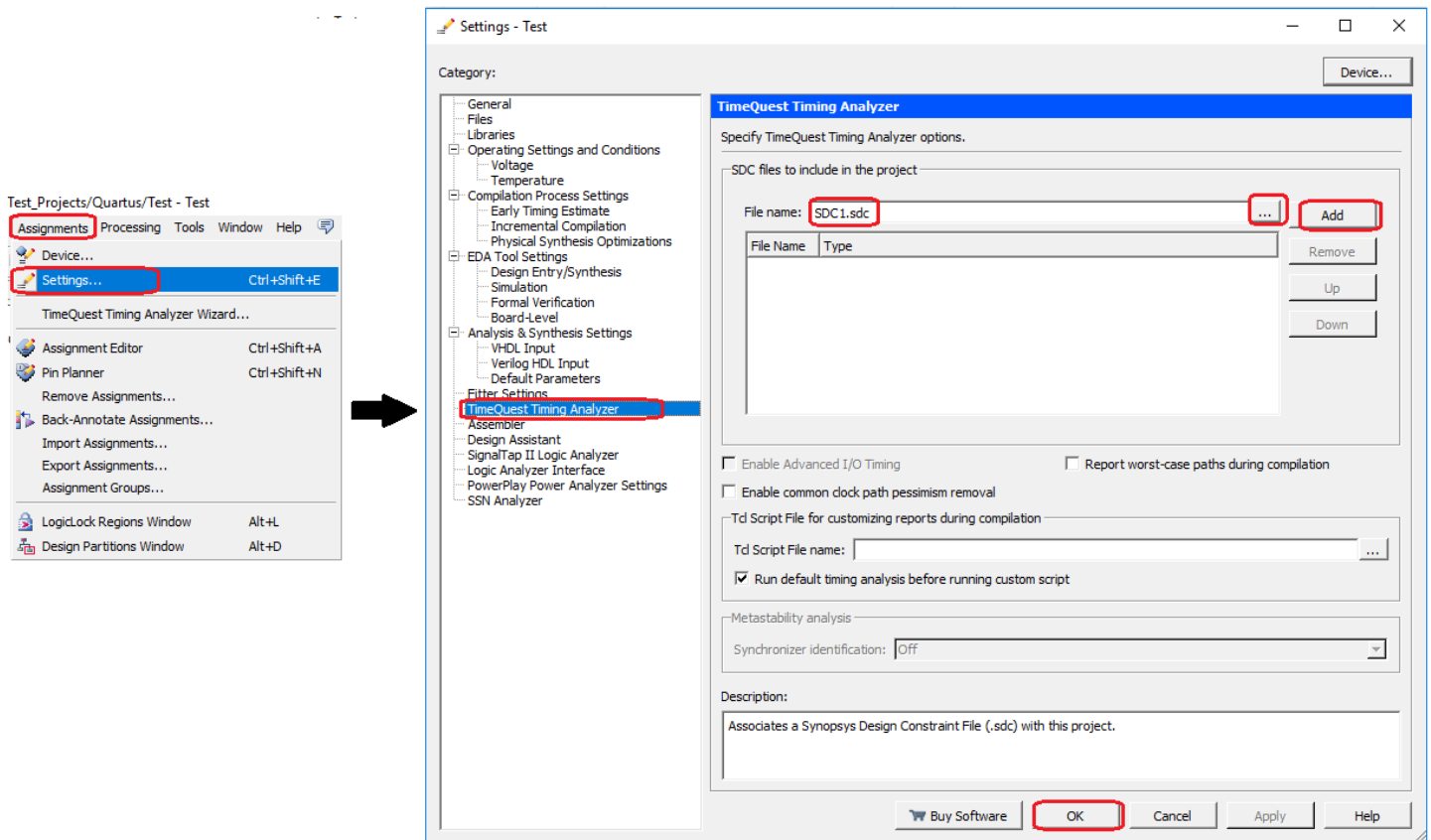


5. Step 5 – Setting system constraints:

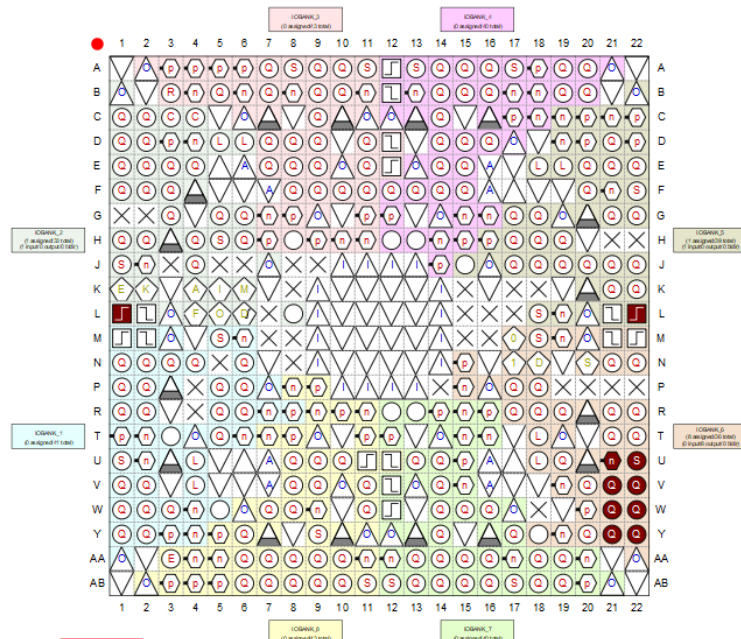
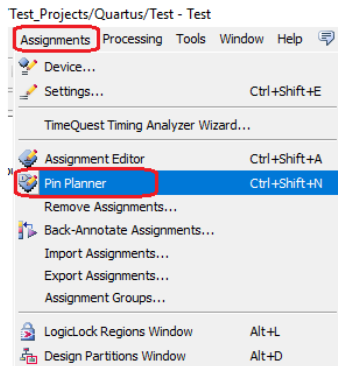
a. Create the system constraints *.sdc file:



b. Add the *.sdc file to the project:



6. Step 6 – Pin Planner Layout (See DE1-Standard user manual on pages 24-26):



Node Name	Direction	Location	I/O Bank	VREF Group	Filter Location	I/O Standard	Reserved	Current Strength	Slew Rate
clk	Input	PIN_Y2	2	B2_N0	PIN_J1	2.5 V (default)		8mA (default)	
enable	Input	PIN_AB28	5	B5_N1	PIN_J4	2.5 V (default)		8mA (default)	
q[7]	Output	PIN_H19	7	B7_N2	PIN_K2	2.5 V (default)		8mA (default)	2 (default)
q[6]	Output	PIN_J19	7	B7_N2	PIN_L3	2.5 V (default)		8mA (default)	2 (default)
q[5]	Output	PIN_E18	7	B7_N1	PIN_M4	2.5 V (default)		8mA (default)	2 (default)
q[4]	Output	PIN_F18	7	B7_N1	PIN_K3	2.5 V (default)		8mA (default)	2 (default)
q[3]	Output	PIN_F21	7	B7_N0	PIN_G2	2.5 V (default)		8mA (default)	2 (default)
q[2]	Output	PIN_E19	7	B7_N0	PIN_K4	2.5 V (default)		8mA (default)	2 (default)
q[1]	Output	PIN_F19	7	B7_N0	PIN_G1	2.5 V (default)		8mA (default)	2 (default)
q[0]	Output	PIN_G19	7	B7_N2	PIN_K1	2.5 V (default)		8mA (default)	2 (default)

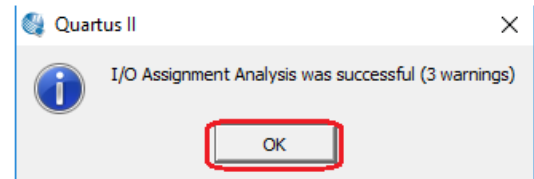
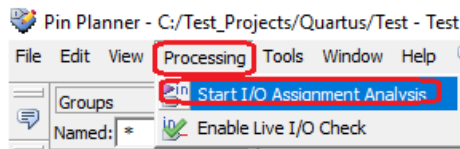


Table 4-5 Pin Assignments for Clock Inputs

Signal Name	FPGA Pin No.	Description	I/O Standard
CLOCK_50	PIN_Y2	50 MHz clock input	3.3V

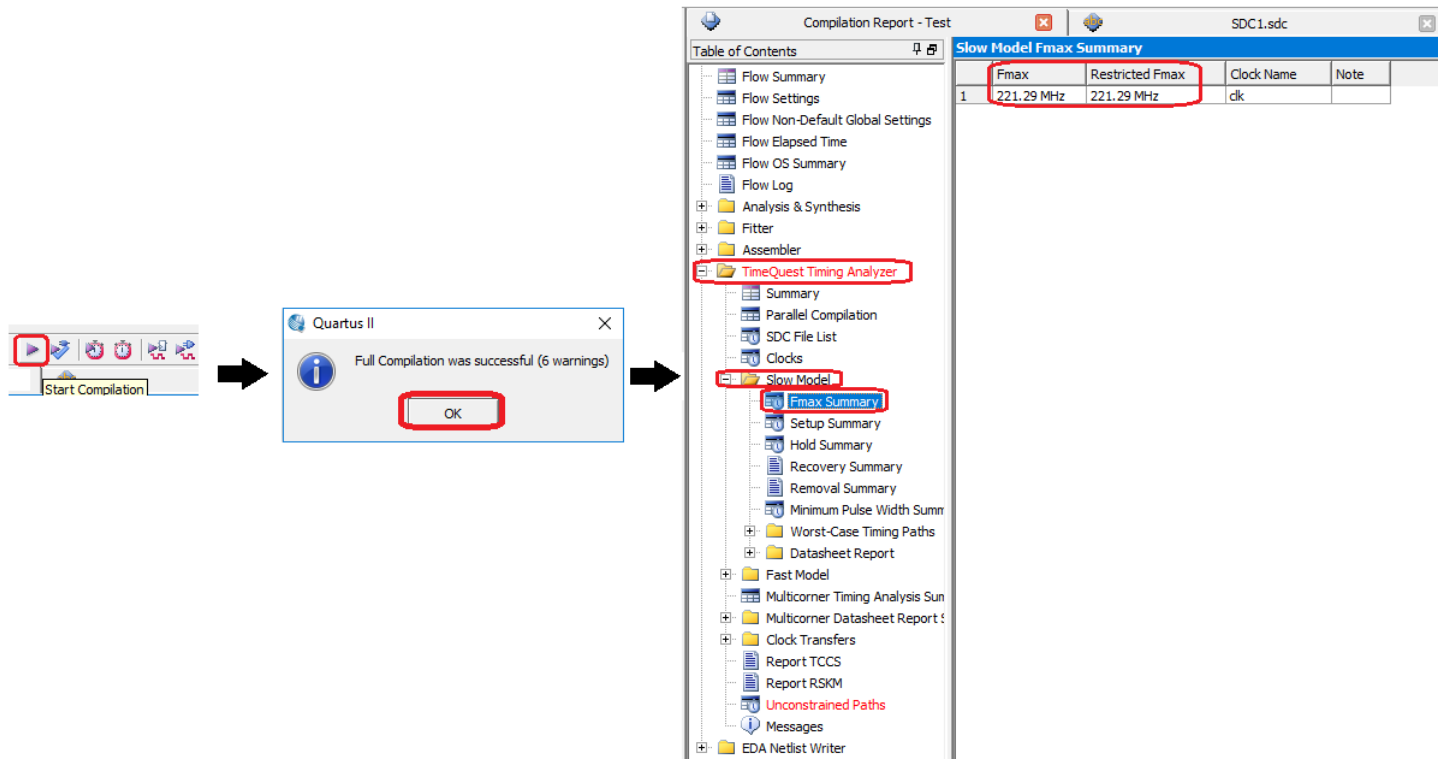
Table 4-1 Pin Assignments for Slide Switches

Signal Name	FPGA Pin No.	Description	I/O Standard
SW[0]	PIN_AB28	Slide Switch[0]	Depending on JP7

Table 4-3 Pin Assignments for LEDs

Signal Name	FPGA Pin No.	Description	I/O Standard
LEDR[0]	PIN_G19	LED Red[0]	2.5V
LEDR[1]	PIN_F19	LED Red[1]	2.5V
LEDR[2]	PIN_E19	LED Red[2]	2.5V
LEDR[3]	PIN_F21	LED Red[3]	2.5V
LEDR[4]	PIN_F18	LED Red[4]	2.5V
LEDR[5]	PIN_E18	LED Red[5]	2.5V
LEDR[6]	PIN_J19	LED Red[6]	2.5V
LEDR[7]	PIN_H19	LED Red[7]	2.5V

7. Step 7 – Full compilation (finding of f_{max}):



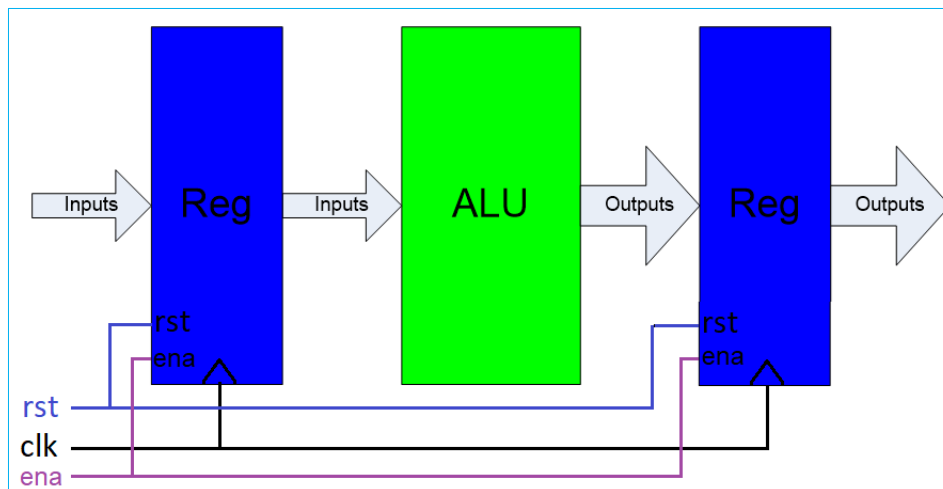
Compilation Report - Test SDC1.sdc

Table of Contents

- Flow Summary
- Flow Settings
- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Fitter
- Assembler
- TimeQuest Timing Analyzer
- Summary
- Parallel Compilation
- SDC File List
- Clocks
- Slow Model
- Fmax Summary
- Setup Summary
- Hold Summary
- Recovery Summary
- Removal Summary
- Minimum Pulse Width Summ
- Worst-Case Timing Paths
- Datasheet Report
- Fast Model
- Multicorner Timing Analysis Sun
- Multicorner Datasheet Report !
- Clock Transfers
- Report TCCS
- Report RSKM
- Unconstrained Paths
- Messages
- EDA Netlist Writer

Slow Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	221.29 MHz	221.29 MHz	clk	



Explanation: In order Quartus IDE can calculate f_{max} the logic design parts must be wrapped by registers

8. Step 8 – Finding Critical Path Location:

The screenshot displays the TimeQuest Timing Analyzer interface. On the left, the **Tools** menu is open, with **Timing Analyzer** highlighted. A large black arrow points from this menu item to the **Tasks** pane in the center. The **Tasks** pane shows a tree of reports, with **Report Timing...** selected and highlighted by a red box. A red text label "Mouse right click" points to this item. A context menu is open over **Report Timing...**, showing **Start** and **Start Again** options, with **Start** highlighted by a red box. The **Report** pane at the top shows "Report not available". The **Console** window at the bottom displays Tcl commands and their output, including `project_open -force "C:/Test_Projects/Quartus/Test.qpf" -revision T`.

Tools Window Help

- Run Simulation Tool
- Launch Simulation Library Compiler
- Launch Design Space Explorer II
- Timing Analyzer**
- Advisors
- Chip Planner
- Design Partition Planner
- Netlist Viewers

Tasks

- Report Setup Summary
- Report Hold Summary
- Report Recovery Summary
- Report Removal Summary
- Report Minimum Pulse Width Summary
- Report Max Skew Summary
- Datasheet
 - Report Fmax Summary
 - Report Datasheet
- Device Specific
 - Report TCCS
 - Report RSKM
 - Report DDR
 - Report Metastability
- Diagnostics
 - Report Clocks
 - Report Clock Transfers
 - Report Unconstrained Paths
 - Report SDC
 - Report Ignored Constraints
 - Check Timing
 - Report Partitions
- Custom Reports
 - Report Timing...**
 - Report Minimum P...
 - Report False Path...
 - Report Path...
 - Report Exceptions...
 - Report Bottlenecks...
 - Report Net Timing...
 - Report Skew...
 - Report Max Skew...

Report Timing... Start Start Again

Console

```
assumed to be external and are run using Tcl's "exec"
command.
- Type "exit" to exit.
- Type "help" to view a list of Quartus II Tcl packages.
- Type "help <package name>" to view a list of Tcl commands
available for the specified Quartus II Tcl package.
- Type "help -tcl" to get an overview on Quartus II Tcl usages.
*****
tcl> project_open -force "C:/Test_Projects/Quartus/Test.qpf" -revision T
tcl>
```

Report Timing

Clocks
 From clock:
 To clock:

Targets
 From:
 Through:
 To:

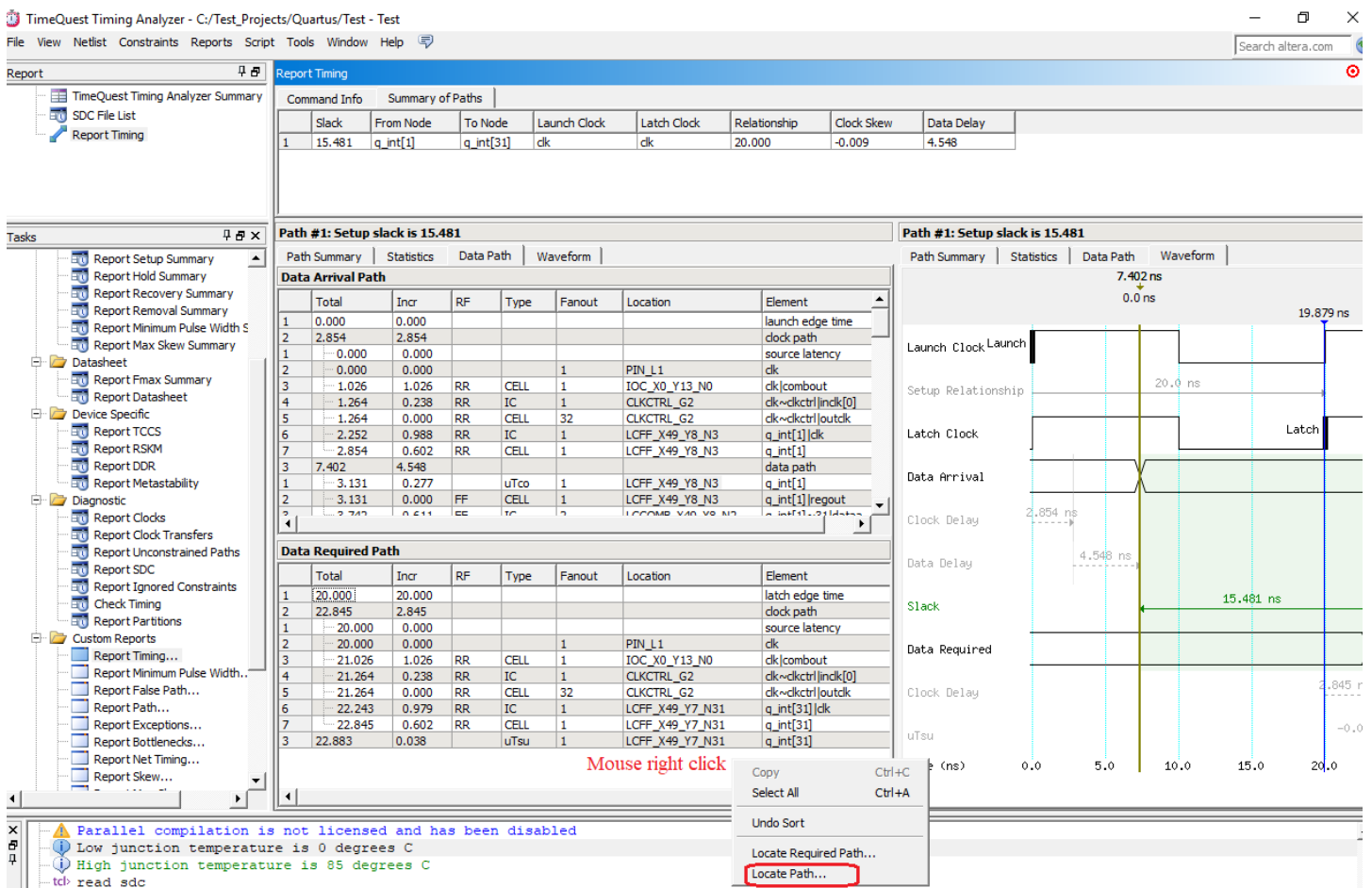
Analysis type
☒ Setup
☐ Hold
☐ Recovery
☐ Removal

Paths
 Report number of paths: 1
 Maximum number of paths per endpoint:
 Maximum slack limit: ns
☐ Pairs only

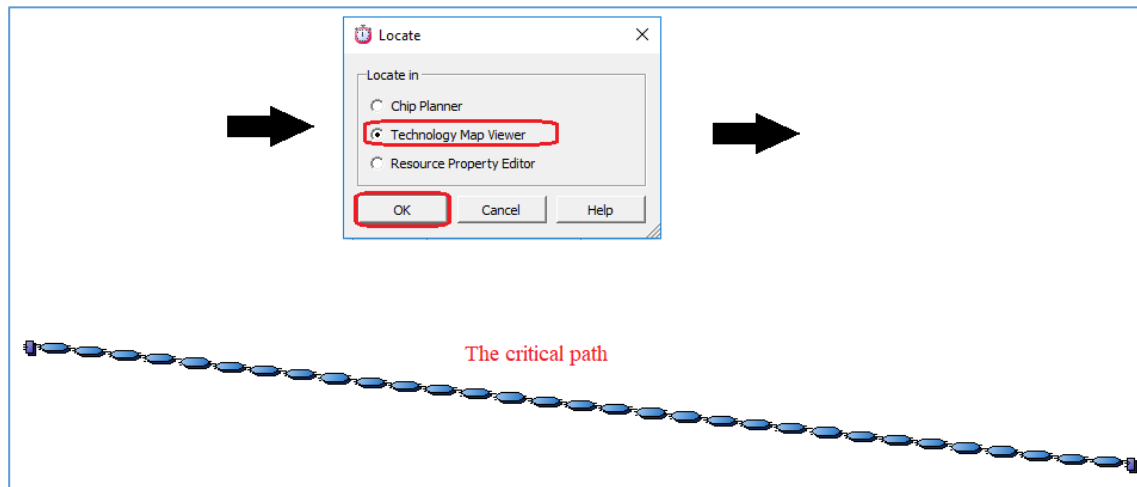
Output
 Detail level: Full path Set Default
☐ Show routing
☒ Report panel name: Report Timing
☐ File name:
 File options
☒ Overwrite ☐ Append Open
☐ Console

Td command: report_timing -setup -npaths 1 -detail full_path -panel_name {Report Timing}

Report Timing Close Help



i. Technology Map Viewer:



ii. Resource Property Editor:

The 'Locate' dialog box is shown with the 'Resource Property Editor' option selected. Below the dialog, the Resource Property Editor window is displayed, showing the properties of the selected resource.

Node Selection Table:

Node Name	Location
☒ [counter]q_int[10]~49	LCCOMB_X49_Y8_N20
☐ [counter]q_int[11]~51	LCCOMB_X49_Y8_N22
☐ [counter]q_int[12]~53	LCCOMB_X49_Y8_N24
☐ [counter]q_int[13]~55	LCCOMB_X49_Y8_N26
☐ [counter]q_int[14]~57	LCCOMB_X49_Y8_N28
☐ [counter]q_int[15]~59	LCCOMB_X49_Y8_N30
☐ [counter]q_int[16]~61	LCCOMB_X49_Y7_N0
☐ [counter]q_int[17]~63	LCCOMB_X49_Y7_N2
☐ [counter]q_int[18]~65	LCCOMB_X49_Y7_N4
☐ [counter]q_int[19]~67	LCCOMB_X49_Y7_N6
☐ [counter]q_int[1]	LGFF_X49_Y8_N3
☐ [counter]q_int[1]~31	LCCOMB_X49_Y8_N2
☐ [counter]q_int[20]~69	LCCOMB_X49_Y7_N8
☐ [counter]q_int[21]~71	LCCOMB_X49_Y7_N10

Properties/Modes Table:

Properties/Modes	Values
Sum LUT Mask	SA5F
Carry LUT Mask	SF5F
Operation Mode	arithmetic
Latch Type	none
Sum Equation	A & IC # 1A & (C # 1D)
Carry Equation	1C # 1A

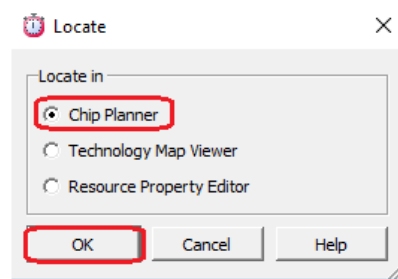
Input Port Name Table:

Input Port Name	Signal Name	Inverted
Register		
EN	[counter]en	False
SCLR	<Disconnected>	False
IACLR	<Disconnected>	False
SDATA	<Disconnected>	False
DATAIN	[counter]q_int[10]~49	False
SLOAD	<Disconnected>	False
CLK	[counter]clk~clkctrl	False
Combinational		
DATA	<Disconnected>	False
DATAA	[counter]q_int[10]	False
DATAB	<Disconnected>	False
DATAD	VCC	False
CTN	[counter]q_int[9]~48	False

Output Port Name Table:

Output Port Name	Signal Name
Register	
REGOUT	[counter]q_int[10]
Combinational	
COUT	[counter]q_int[10]~50
COMBOUT	[counter]q_int[10]~49

iii. Chip Planner:



Coordinate: Editing Mode: ECO - EP2C20F484C7

Layers Settings

Basic

- ☒ Background
 - ☐ None
 - ☒ Block Utilization
 - ☐ Design Partition P...
- ☒ LogicLock Regions
 - ☒ User-assigned Lo...
 - ☒ Fitter-placed Logi...
- ☐ Clock Regions
 - ☐ Global Clock Region
 - ☐ Local Clock Region
 - ☐ LVDS Clock Region
 - ☐ Regional Clock R...
 - ☐ Periphery Clock ...
- ☒ Overlay Objects
 - ☒ Connection Lines
 - ☒ Labels
 - ☒ Differential Pin P...
 - ☒ Report Overlay

Layers Settings Color Legend

Properties

Selected elements:

General

Timing

Located 1 paths

Time	Object
15.481	q_int[1] -> q_int[31]
1.026ns	clk -> q_int[1]
0.238ns	clk -> clk~clkctrl
0.000ns	clk~clkctrl -> clk~clkctrl
0.988ns	clk~clkctrl -> q_int[1]
0.602ns	q_int[1] -> q_int[31]
Arrival Data	q_int[1] -> q_int[31]
(Required Clock)	clk -> q_int[31]

9. Step 9 – Quartus Power Analyzer:

Quartus power analysis using a *.vcd (Value Change Dump) file provides accurate power estimation by analyzing real signal activities from simulation. The process involves generating a VCD file from ModelSim during gate-level simulation, configuring the Power Analyzer in Quartus, and running the analysis.

Steps for Power Analysis Using a VCD File:

a. Generate the VCD File:

- i. Compile-All and start the simulation of the required ModelSim project:

Tcl commands:

project calculateorder

project compileall

vsim tb_entity_name

- ii. Create a *.vcd file, add all DUT signals, run the simulation, and save the file:

Tcl commands:

*vcd file filename.vcd // create a *.vcd file*

vcd add tb_entity_name / // add the top entity signals (keep tb and top the same signal names)*

run 20 us // in case the clock cycle time is 20ns, run simulation for 1000 cycles

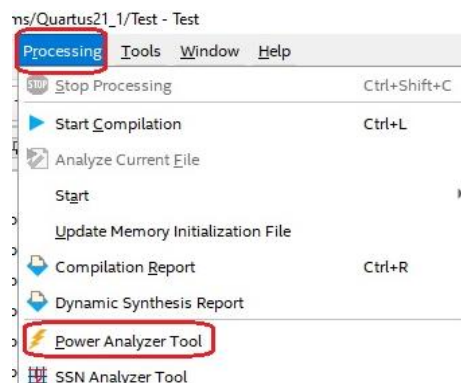
vcd flush // immediate writing of buffered VCD data to the VCD file

b. Configure Quartus Power Analyzer:

- i. Open your project in Quartus
- ii. Go to **Assignments > Settings > Power Analyzer Settings**
- iii. Under **Input signal activities**, select **Simulation Results**
- iv. Select **File name** and browse to your generated .vcd file, click OK
- v. Click **Processing > Power Analyzer Tool > Set Operating Conditions** (cooling Solution and Temperature ...)
- vi. Click **Start**, and at the simulation's end, click **Report** to get the Power Analysis report

Key Tips for VCD Power Analysis:

- i. **Glitch Filtering:** Use the recommended simulator settings to enable glitch filtering to prevent overestimating dynamic power.
- ii. **Vectorless Estimation:** If you do not have a full VCD file, you can use vectorless estimation, but it is less accurate.
- iii. **Simulation Time:** Ensure the VCD file captures a long enough period to represent typical design activity.



Compilation Report - Test X Power Analyzer Tool X

Input file

☒ Use input file(s) to initialize toggle rates and static probabilities during power analysis

Add Power Input File(s)... **Step 1: Insert the VCD file**

Output file

☒ Write out signal activities used during power analysis

Output file name:

Early Power Estimation File

☒ Write out Early Power Estimation file

Output file name:

Default toggle rates for unspecified signals

Default toggle rate used for input I/O signals: %

Default toggle rate used for remaining signals

☐ Use default value: %

☒ Use vectorless estimation

Step 2

0%

00:00:00

Start **Step 3** **Stop** **Report** **Step 4**

Settings - Test

VCD file insertion properties

Category:

- General
- Files
- Libraries
- IP Settings
 - IP Catalog Search Locations
 - Design Templates
- Operating Settings and Conditions
 - Voltage
 - Temperature
- Compilation Process Settings
 - Incremental Compilation
- EDA Tool Settings
 - Design Entry/Synthesis
 - Simulation
 - Board-Level
- Compiler Settings
 - VHDL Input
 - Verilog HDL Input
 - Default Parameters
 - Timing Analyzer
 - Assembler
 - Design Assistant
 - Signal Tap Logic Analyzer
 - Logic Analyzer Interface
 - Power Analyzer Settings
 - SSN Analyzer

Power Analyzer Settings

Select the power analyzer options.

☒ Run Power Analyzer during compilation

☒ Use input file(s) to initialize toggle rates and static probabilities during power analysis

Input File(s)

File Name	Type	Entity	VCD Start Time	VCD End Time	
C:\Users\revoh\OneDrive - Ben Gurion University of the Negev\My Domain\Computer Architecture\Materials\FPGA+Quartus\Code...	Value Change Dump File	counter			Add...
					Edit...
					Remove

☒ Perform glitch filtering on VCD files

☒ Write out signal activities used during power analysis

Output file name:

☒ Write signal activities to report file

☒ Write power dissipation by block to report file

Default toggle rates for unspecified signals

Default toggle rate used for input I/O signals: %

Default toggle rate used for remaining signals

☐ Use default value: %

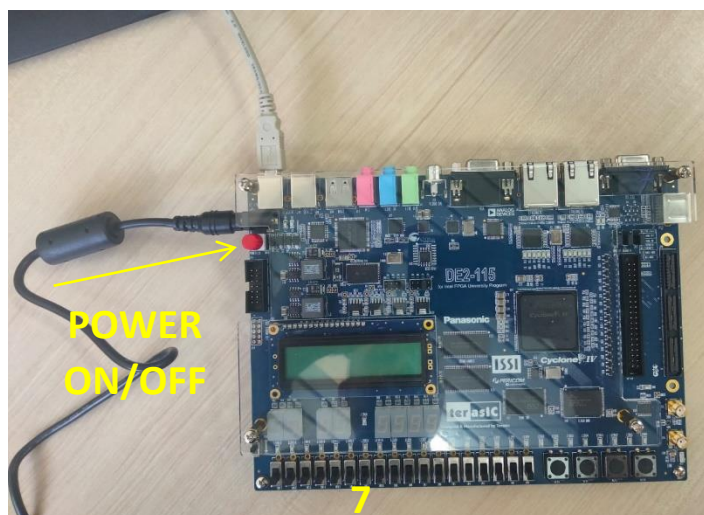
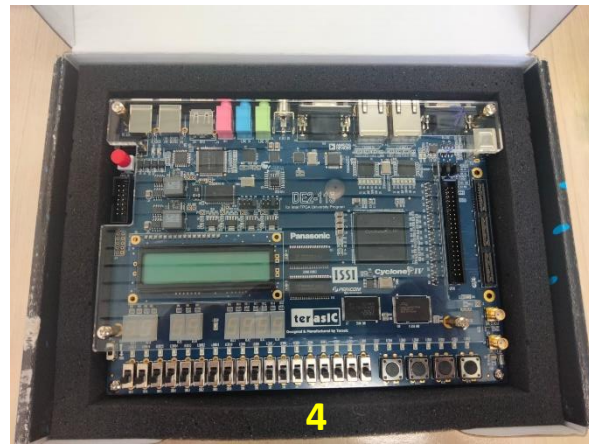
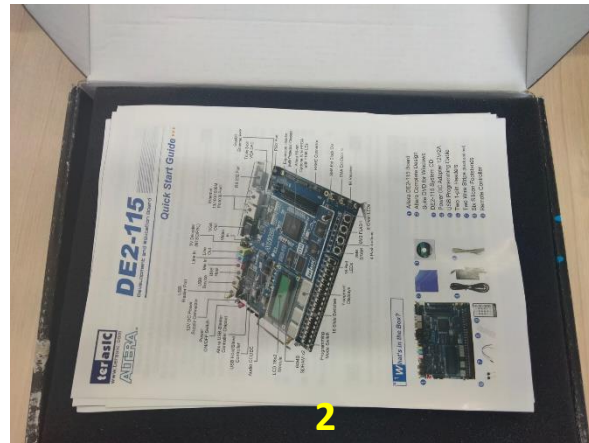
☒ Use vectorless estimation

Buy Software **OK** **Cancel** **Apply** **Help**

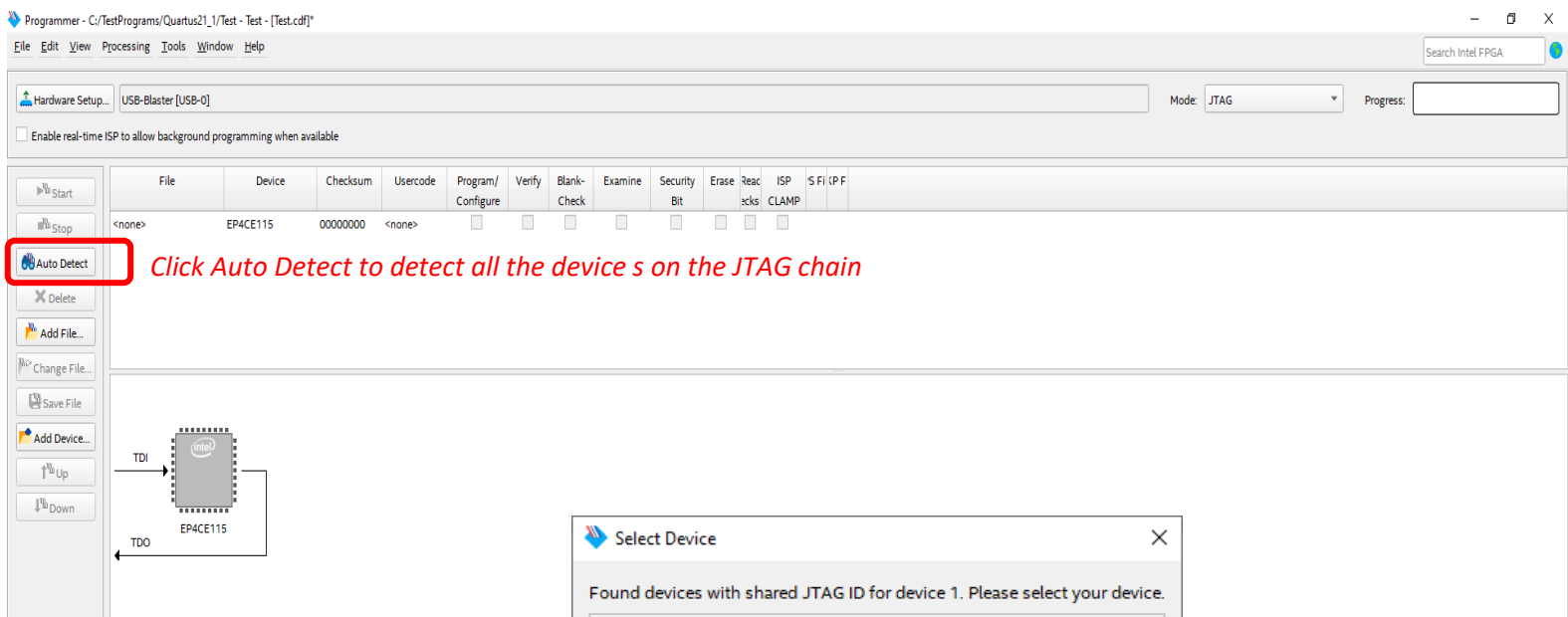
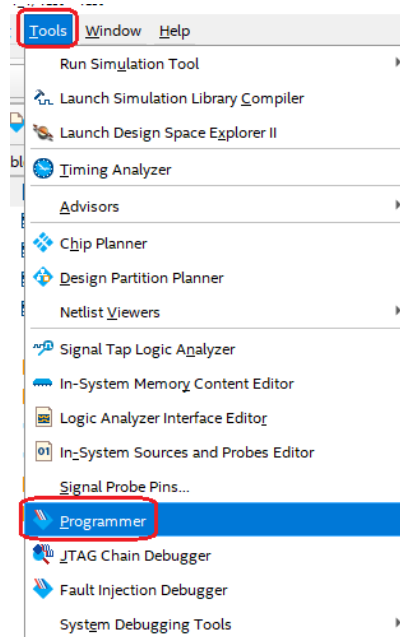


Power Analyzer Summary	
<<Filter>>	
Total Thermal Power Dissipation	245.30 mW
Core Dynamic Thermal Power Dissipation	11.21 mW
Core Static Thermal Power Dissipation	103.40 mW
I/O Thermal Power Dissipation	130.69 mW

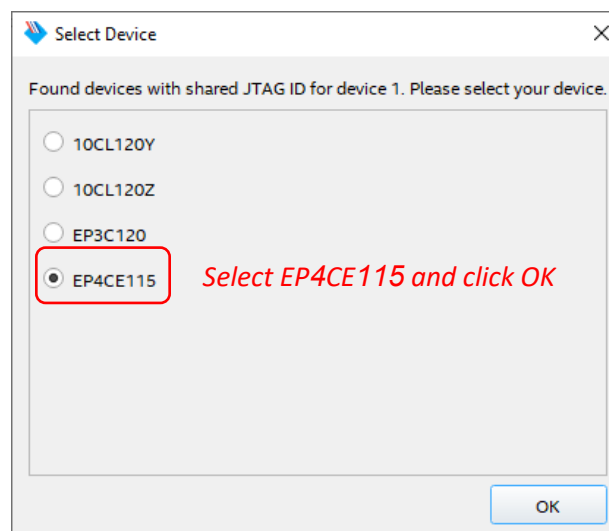
10. Step 10 – Code Programming:



After compiling and verifying your design, you are ready to program the FPGA on the development board. You download the SOF you just created into the FPGA using the USB — BlasterII circuitry on the board. Set up your hardware for programming using the following steps:



Click Auto Detect to detect all the device s on the JTAG chain



Select EP4CE115 and click OK

Programmer - C:/TestPrograms/Quartus21_1/Test - [Test.cdf]*

File Edit View Processing Tools Window Help

Hardware Setup... USB-Blaster [USB-0]

☐ Enable real-time ISP to allow background programming when available

Start Stop Auto Detect Delete Add File... Change File... Save File Add Device... Up Down

File	Device	Checksum	Usercode	Program/Configure	Verify	Blank-Check	Examine	Security Bit	Erase	Reactions	ISP CLAMP	SP	FI	CP	F
output_files/Test.sof	EP4CE115F29	0056789A	0056789A	✓											

Select FPGA the design *.sof file.

Programmer - C:/TestPrograms/Quartus21_1/Test - [Test.cdf]*

File Edit View Processing Tools Window Help

Hardware Setup... USB-Blaster [USB-0] Mode: JTAG Progress: 100% (Successful)

Start Stop Auto Detect Delete Add File... Change File... Save File Add Device... Up Down

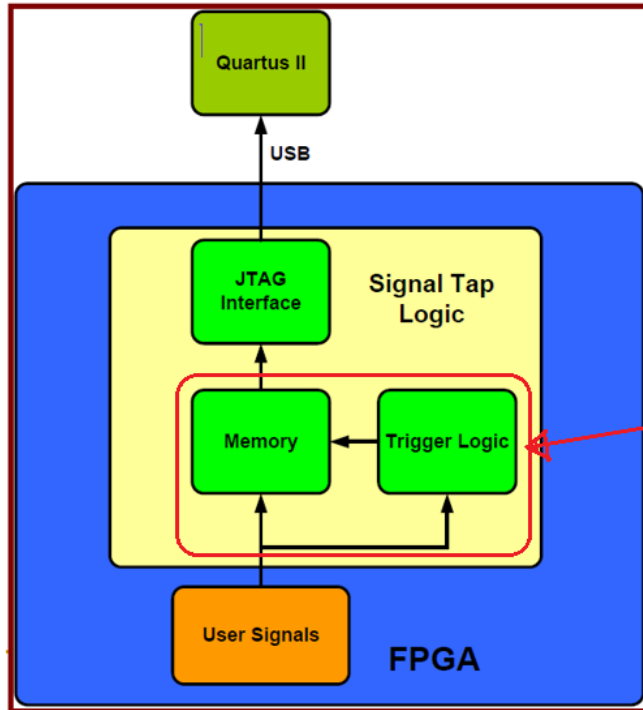
File	Device	Checksum	Usercode	Program/Configure	Verify	Blank-Check	Examine	Security Bit	Erase	Reactions	ISP CLAMP	SP	FI	CP	F
output_files/Test.sof	EP4CE115F29	0056789A	0056789A	✓											

Click Start to program the design *.sof file into FPGA

Congratulations, you have created, compiled, and programmed your first FPGA design! The compiled SRAM Object File (.sof) is loaded onto the FPGA on the development board and the design should be running.*

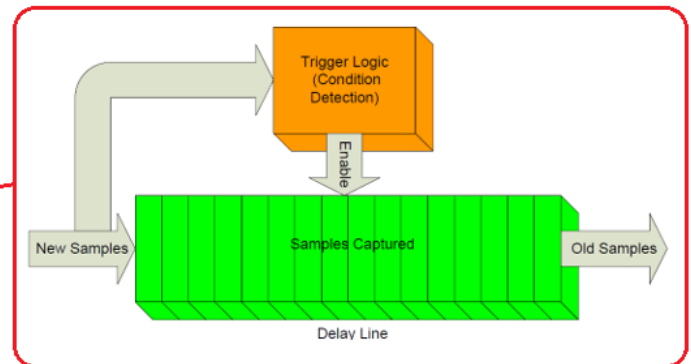
B. Validation – Using Signal TAP:

1. Block Diagram:



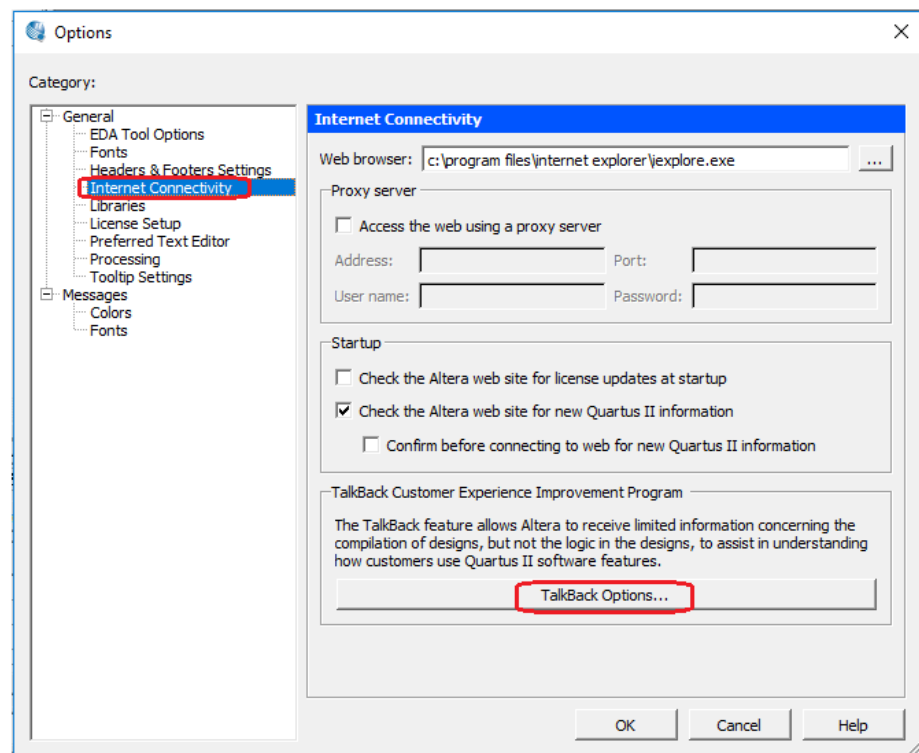
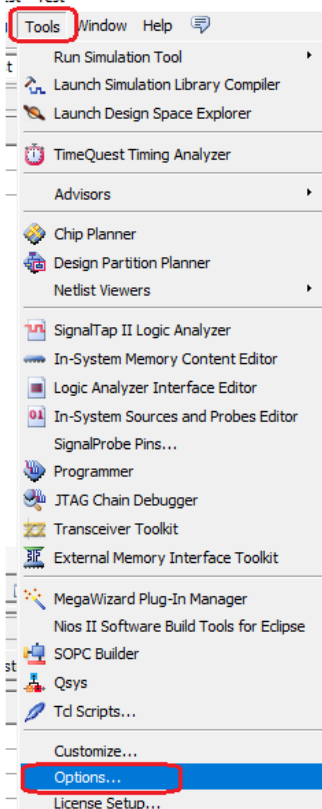
Signal TAP Pros & Cons:

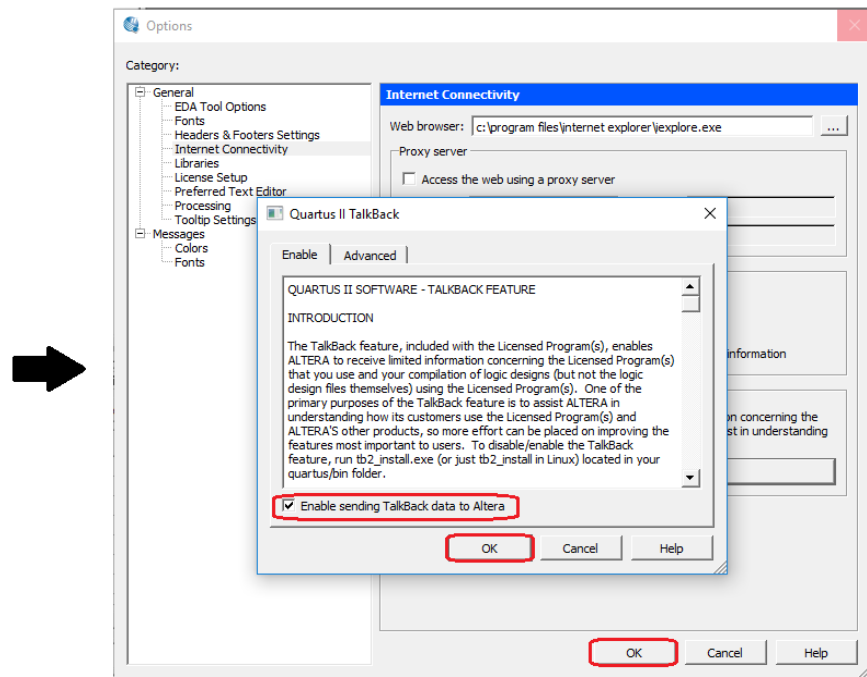
- Captures real time state of FPGA internal signals and pins (up to 200MHz)
- Connects to Quartus II through JTAG
- Do not require huge and expensive equipment
- Uses internal FPGA resources
 - Memory Blocks
 - Logic Elements
- Each time the captured signals list change the design must be recompiled



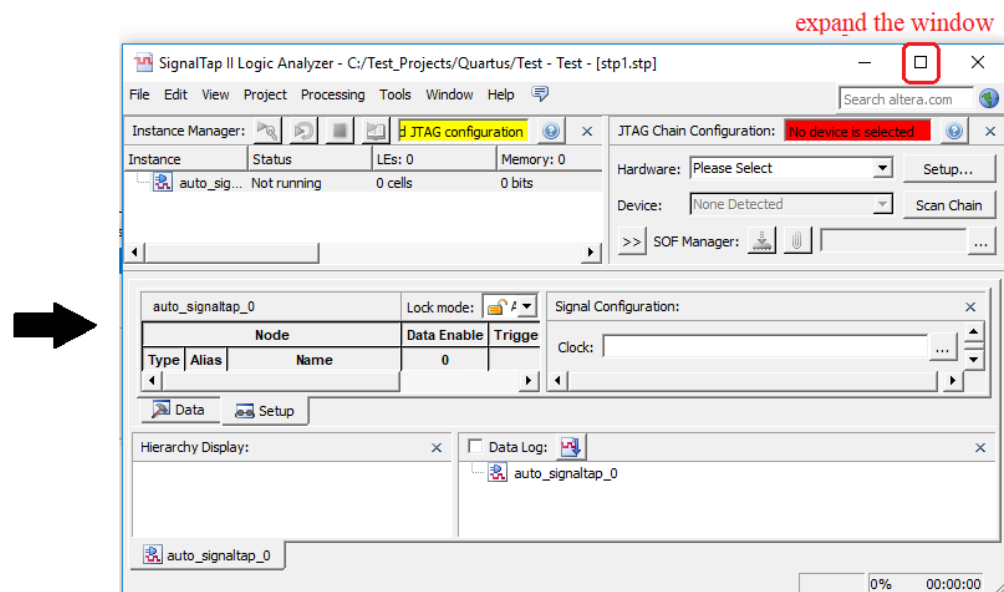
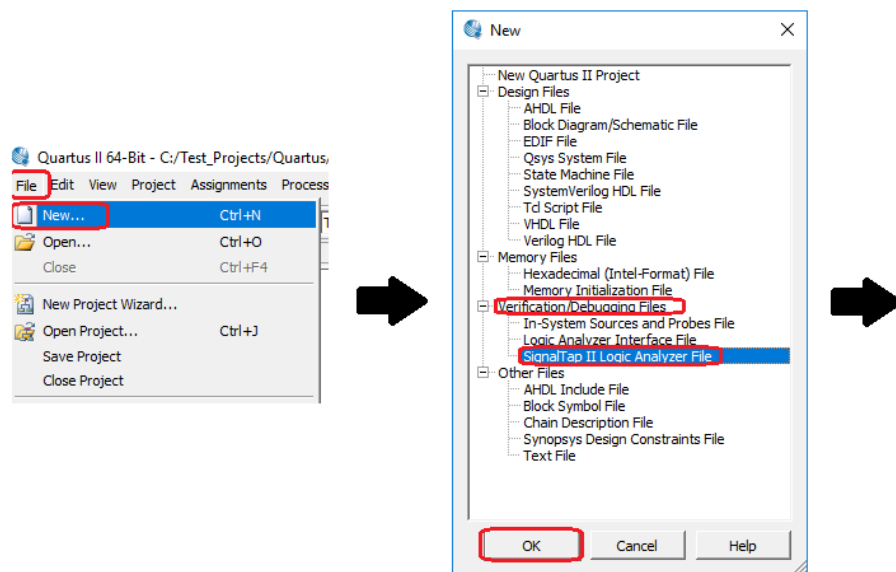
Signal TAP Features:

- Up to 1024 Data Channels
- Multiple Analyzers in One Device
 - Supports Analysis of Multiple Clock Domains
 - Each Analyzer Can Run Simultaneously
- Multiple Analyzers in One Device
- Up to 10 Trigger Levels Per Channel





3. Create a new STP file:

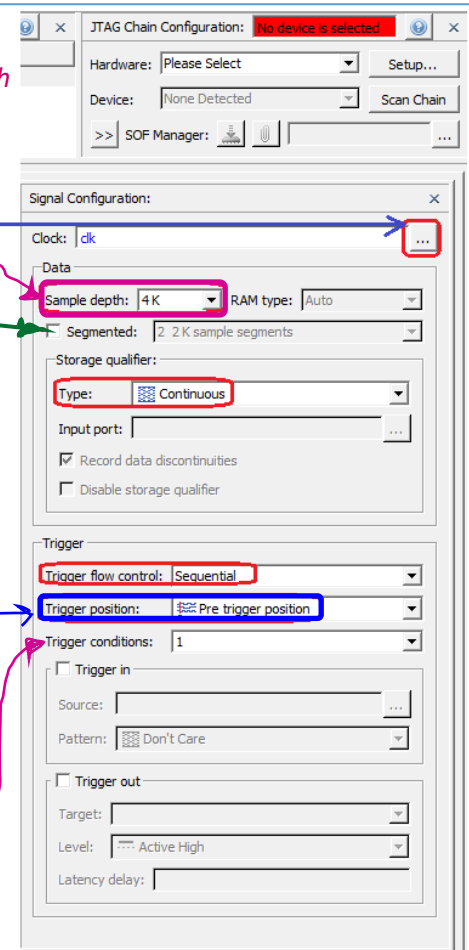
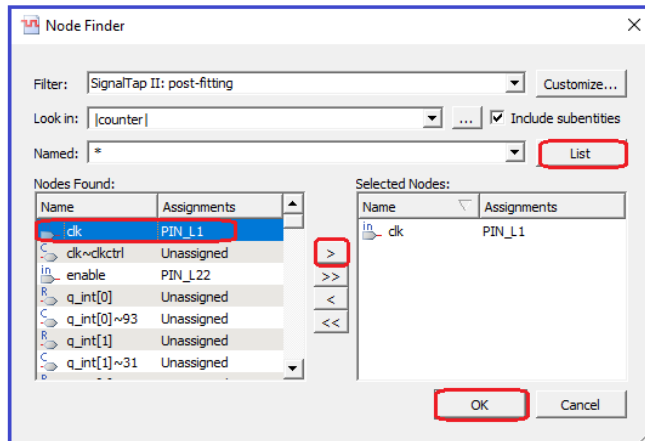


4. Configurations (after expansion of the previous window):

Data segments:

Each segment is associated with a new trigger event. Upon completion of all segments, the results will be shown on the screen.

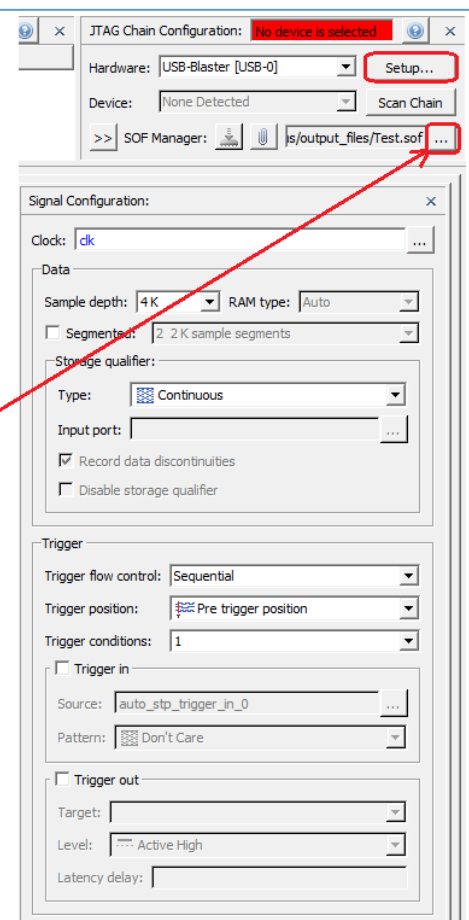
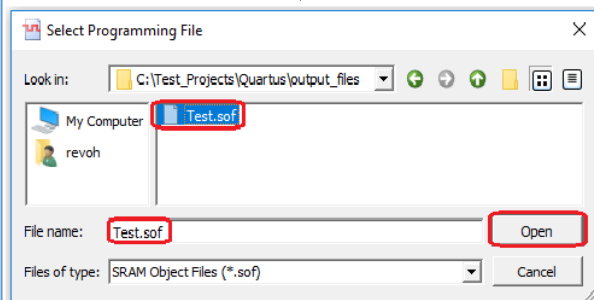
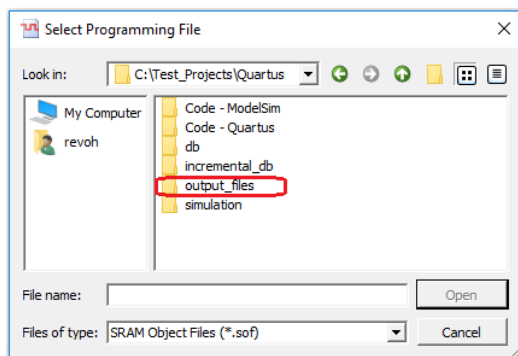
4k samples
(bits) for each
channel line

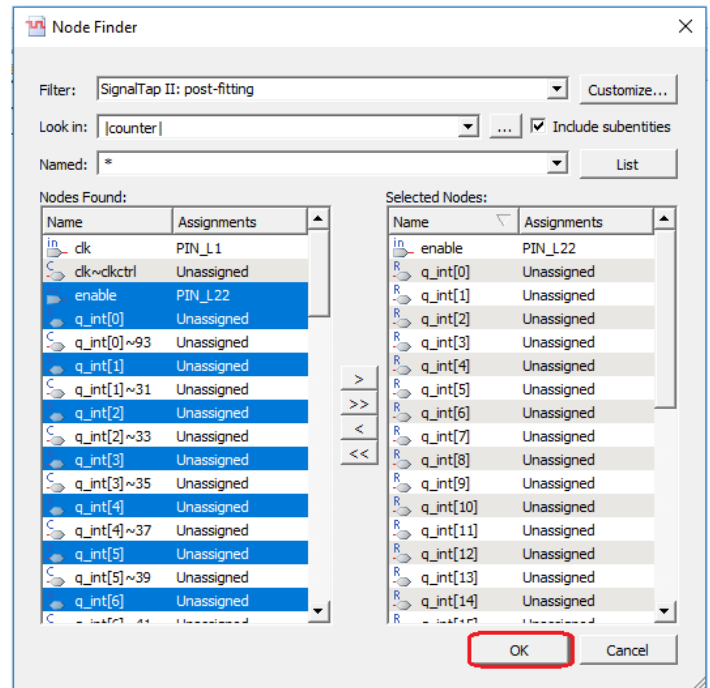
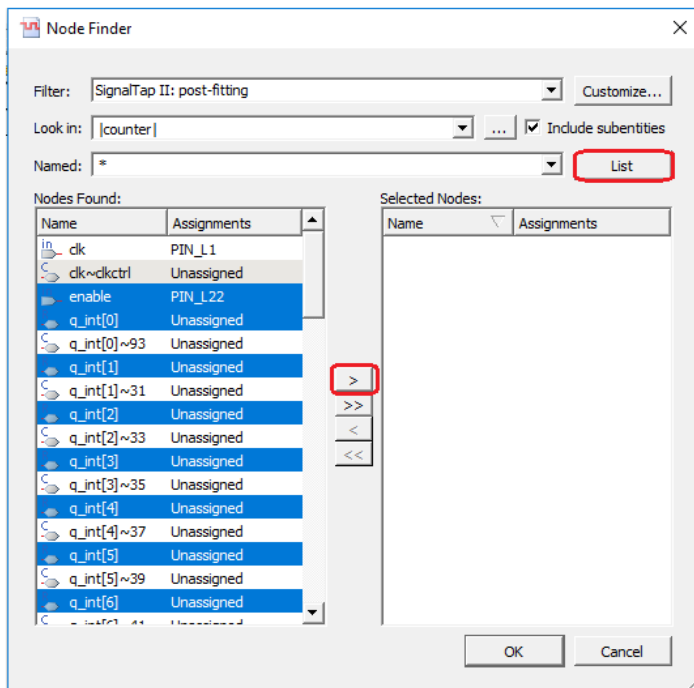
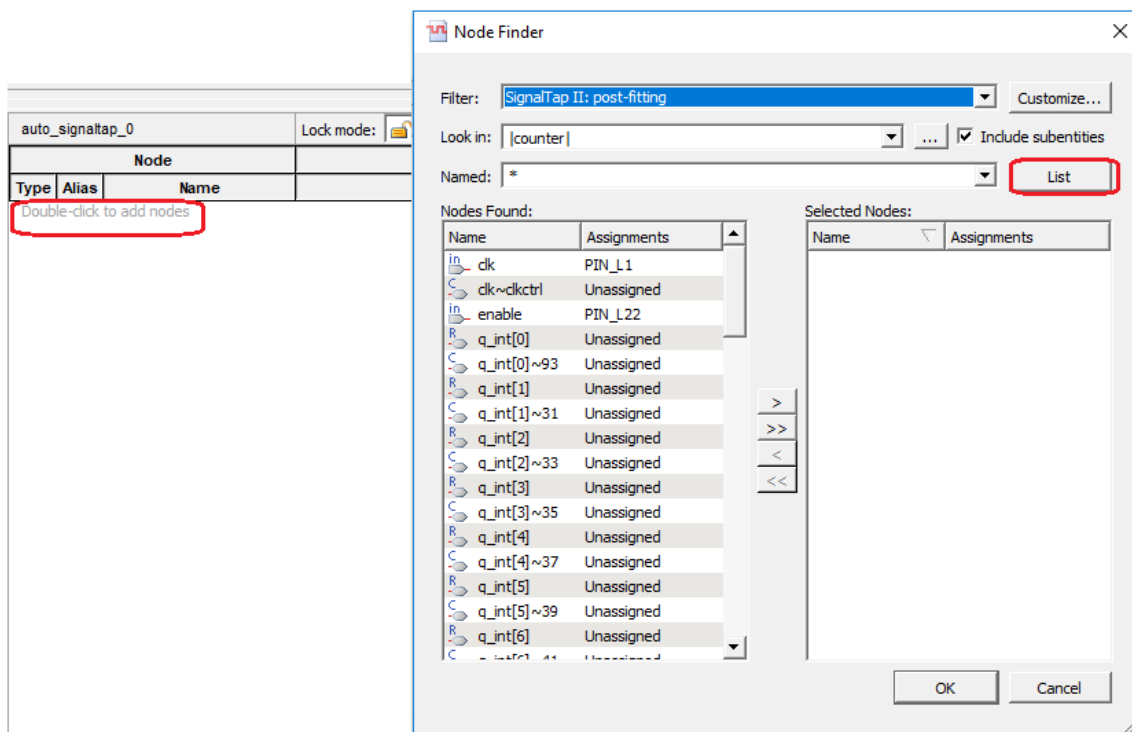


There are three Trigger Position states (circular-buffer based):

1. Pre-Trigger Position:
88% Pre-Trigger + 12% Post-Trigger
2. Center Trigger Position:
50% Pre-Trigger + 50% Post-Trigger
3. Post-Trigger Position:
12% Pre-Trigger + 88% Post-Trigger

The number
of levels of
the Trigger
condition





There is a total of 33 channels, thus the total acquisition memory is $4096 \cdot 33 = 135,168 \text{ bits}$

In case of more than one Trigger condition columns:
1) Between lines in the same column the operation is AND.
2) Between columns the operation is OR.



auto_singaltap_0 Lock mode: Allow all changes

Type	Alias	Name	Data Enable	Trigger Enable	Trigger Conditions
In		enable	33	33	1 Basic
R		q_int[0]			
R		q_int[1]			
R		q_int[2]			
R		q_int[3]			
R		q_int[4]			
R		q_int[5]			
R		q_int[6]			
R		q_int[7]			
R		q_int[8]			
R		q_int[9]			
R		q_int[10]			
R		q_int[11]			
R		q_int[12]			
R		q_int[13]			
R		q_int[14]			
R		q_int[15]			
R		q_int[16]			
R		q_int[17]			
R		q_int[18]			
R		q_int[19]			
R		q_int[20]			
R		q_int[21]			
R		q_int[22]			
R		q_int[23]			
R		q_int[24]			
R		q_int[25]			
R		q_int[26]			

mouse right click

Signal Configuration:

Clock: clk

Sample depth: 4 K

Segmented: 2

Storage qualifier:

Type: Con

Input port:

Record data disc

Disable storage q

Trigger

Trigger flow control: S

Trigger position:

Trigger conditions: 1

Trigger in

Source: auto_stp_

Pattern: Don't C

Trigger out

Target:

Level: Active

Latency delay:

Data Setup



SignalTap II Logic Analyzer - C:/Test_Projec

File Edit View Project Processing Tools V

New File Ctrl+N

Close

Save Ctrl+S

Save As...

Export...

Page Setup...

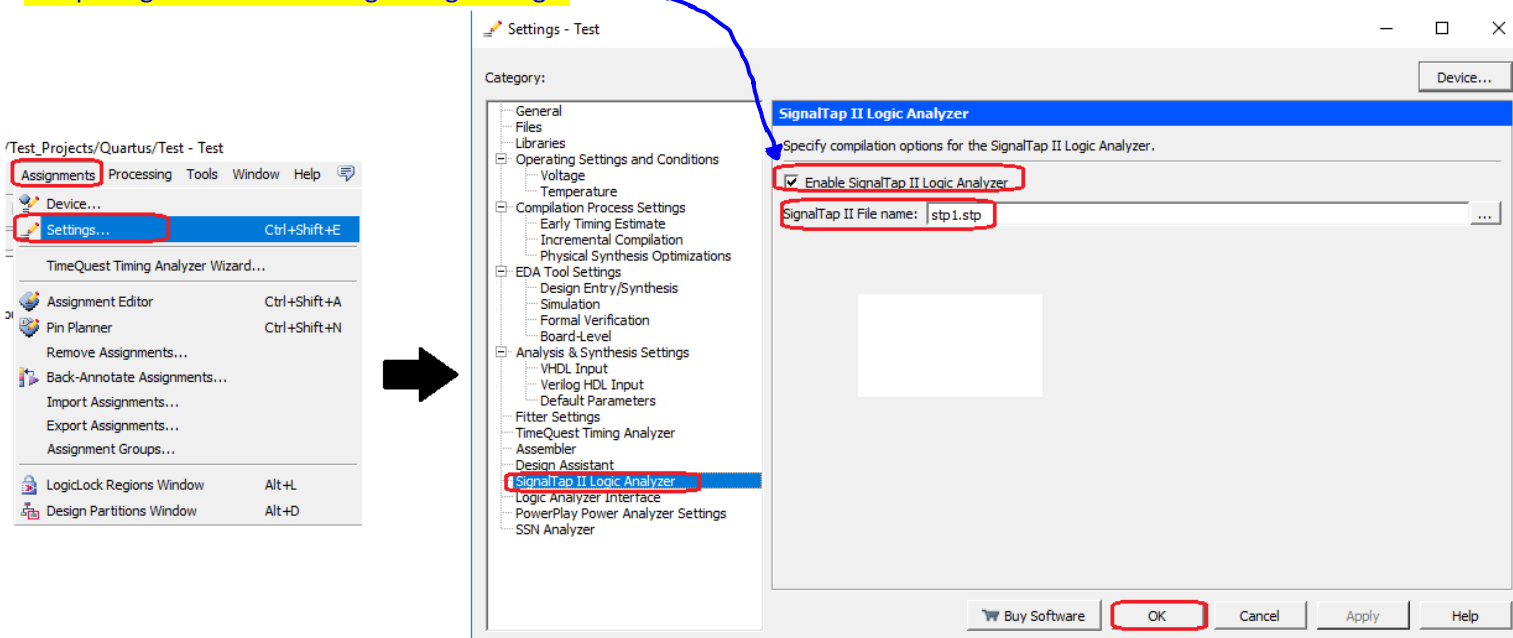
Print Preview

Print... Ctrl+P

Type	Alias	Name
In		enable
R		q_int[0]

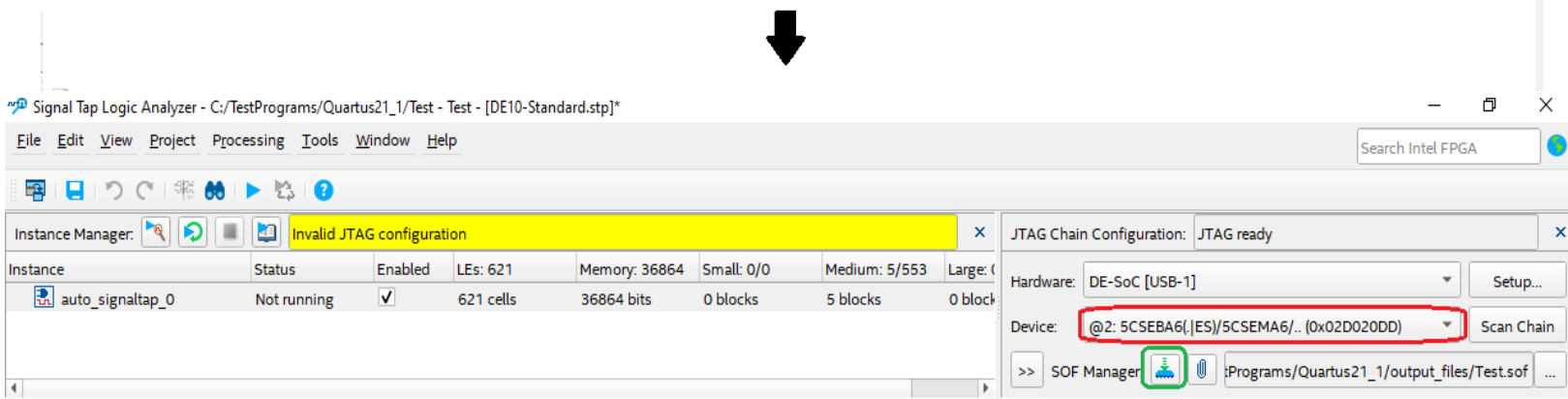
5. Project Compilation and Programming (eventually the Signals results):

Note: After validation using an on-chip embedded logic analyzer (Signal-Tap), remove the *.stp file from the project and compile again for the final digital logic design.



Turn ON the DE1 Board power

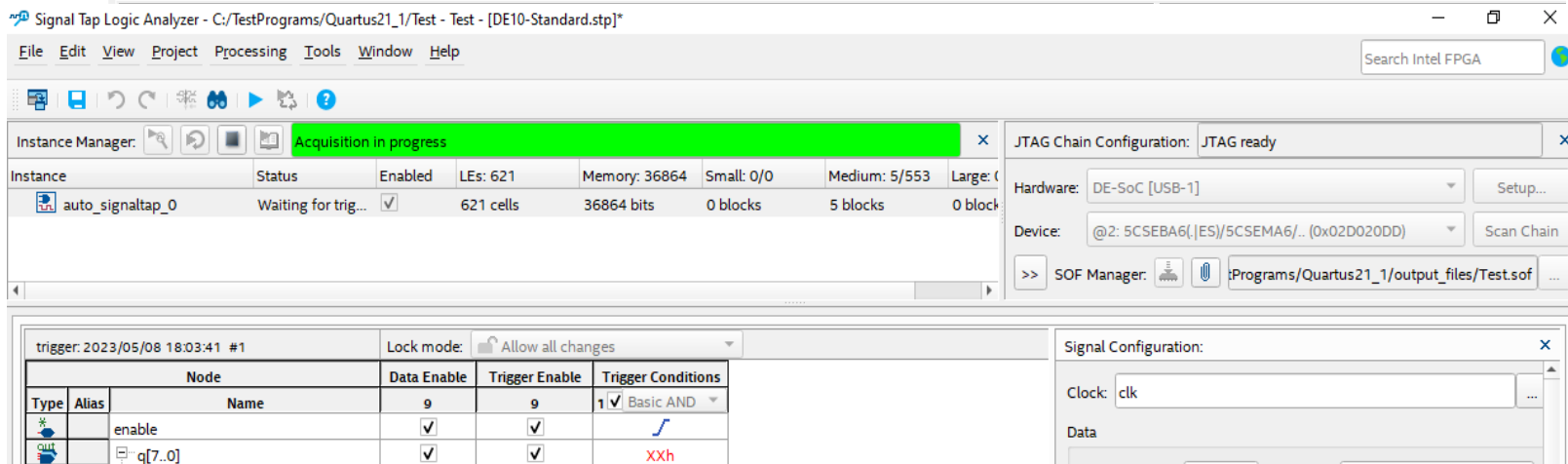
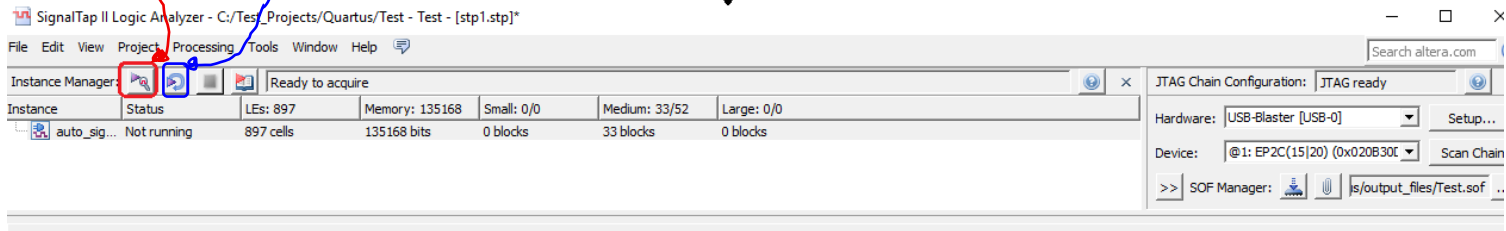
Open the STP file



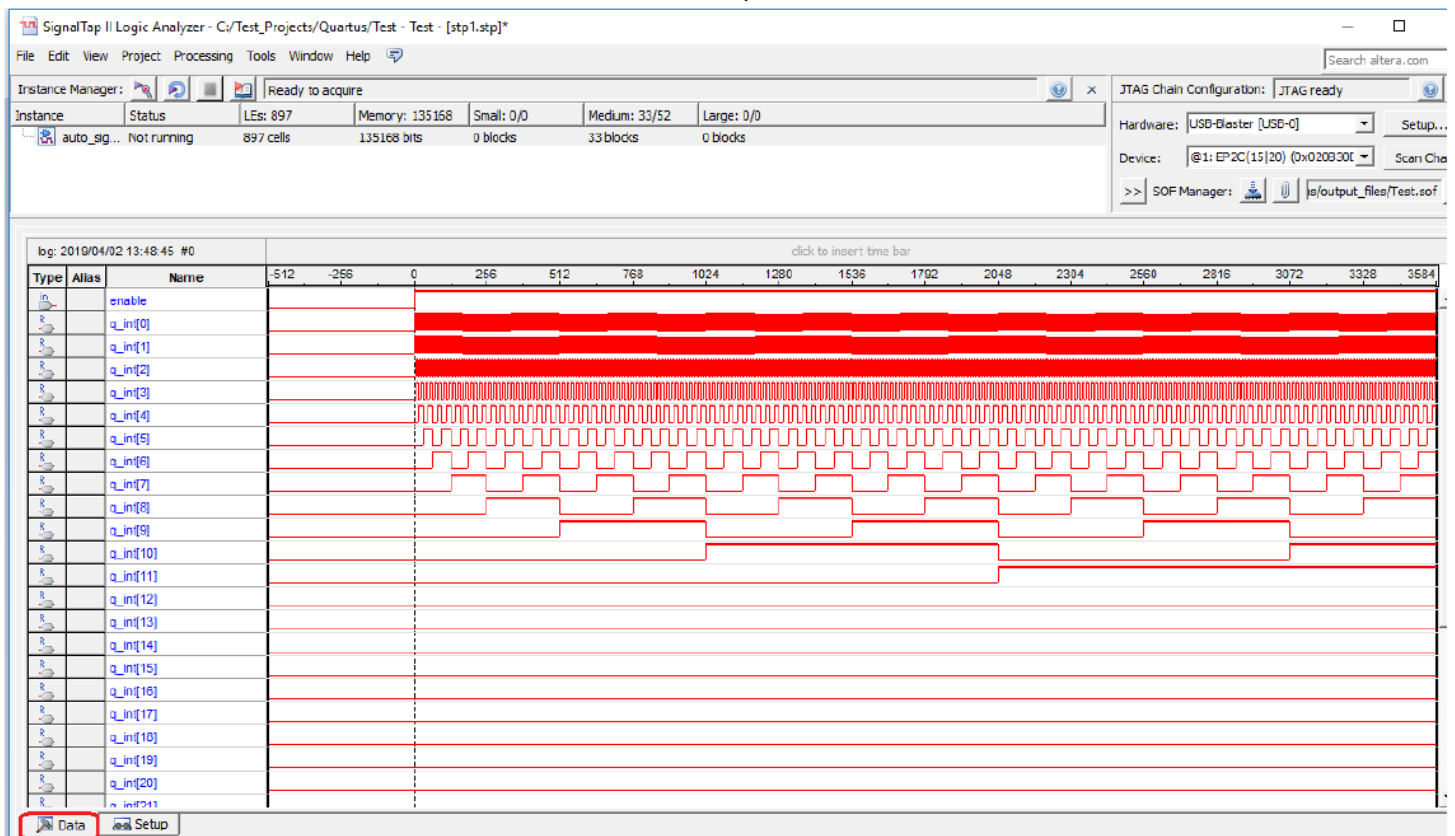
Device programming

Single run for each acquisition progress

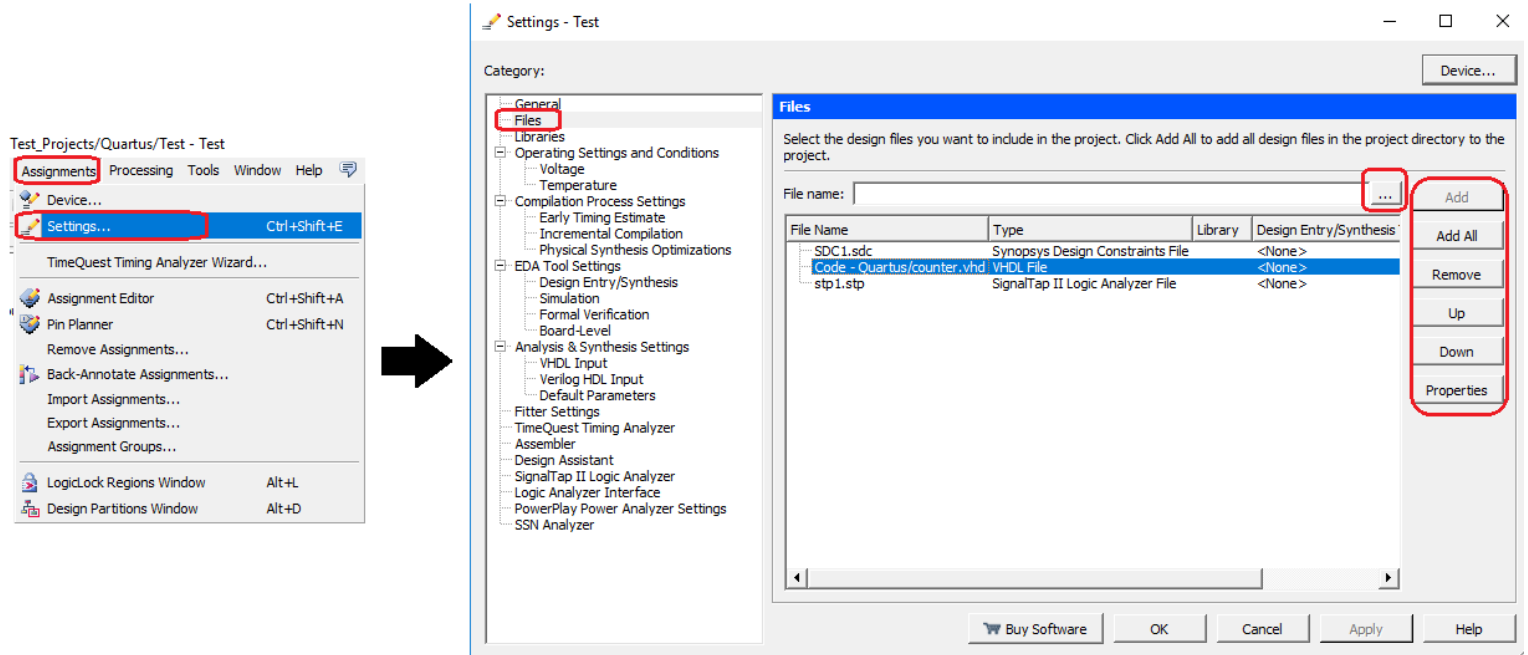
*Auto run, each
trigger event
produces new results*



Assert the Trigger



C. Changing the VHDL source files of the project:



D. Open Existing Project:

